US012396743B2

(12) United States Patent
Martin et al.

(10) Patent No.: US 12,396,743 B2
(45) Date of Patent: Aug. 26, 2025

(54) **RETRIEVAL SYSTEMS AND METHODS FOR USE THEREOF**

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(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 76 days.

(21) Appl. No.: **18/056,199**

(22) Filed: **Nov. 16, 2022**

(65) **Prior Publication Data**

US 2023/0074271 A1 Mar. 9, 2023

Related U.S. Application Data

(63) Continuation of application No. 16/530,101, filed on Aug. 2, 2019, now Pat. No. 11,529,155, which is a continuation of application No. 15/946,466, filed on Apr. 5, 2018, now Pat. No. 11,213,307, which is a continuation of application No. 15/174,016, filed on
(Continued)

(51) **Int. Cl.**

A61B 17/221	(2006.01)
A61B 17/22	(2006.01)
A61B 17/3207	(2006.01)
A61F 2/01	(2006.01)
A61F 2/06	(2013.01)
A61F 2/95	(2013.01)

(Continued)

(52) **U.S. Cl.**
CPC ***A61B 17/221*** (2013.01); ***A61B 17/22031*** (2013.01); ***A61B 17/3207*** (2013.01); ***A61B 17/320758*** (2013.01); ***A61F 2/01*** (2013.01); ***A61F 2/06*** (2013.01); ***A61F 2/95*** (2013.01); ***A61F 2/962*** (2013.01); *A61B 2017/22034* (2013.01); *A61B 2017/3435* (2013.01); *A61B 2560/04* (2013.01); *A61F 2/011* (2020.05); *A61F 2230/0006* (2013.01); *A61F 2230/0093* (2013.01); *A61F 2250/0098* (2013.01)

(58) **Field of Classification Search**
CPC A61B 17/221; A61B 2017/22034; A61F 2230/0006; A61F 2002/018; A61F 2/013; A61F 2230/008; A61F 2002/011; A61F 2/95; A61F 2230/0067; A61F 2/01; A61F 2/962; A61F 2230/0069; A61F 2230/0093; A61F 2002/015
See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

2,918,919	A	12/1959	Wallace
2,943,626	A	7/1960	Enrico

(Continued)

FOREIGN PATENT DOCUMENTS

CN	1640505	A	7/2005
CN	102036611	A	4/2011

(Continued)

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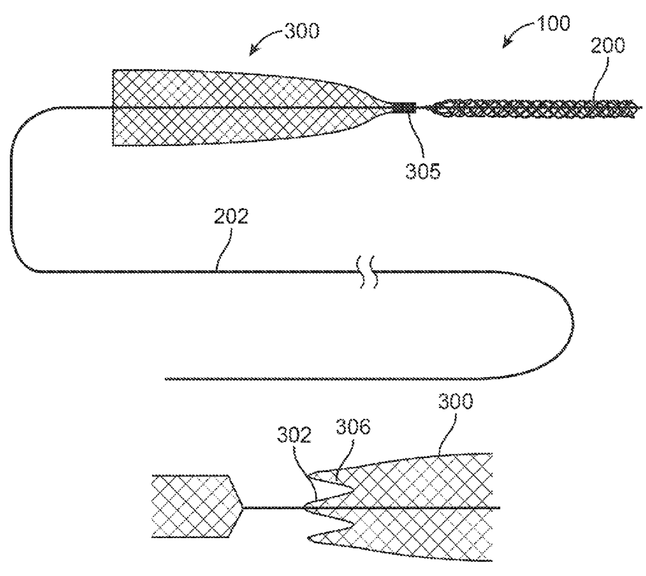
(57) **ABSTRACT**

The devices and methods described herein relate to improved structures for removing obstructions from body lumens. Such devices have applicability in through-out the body, including clearing of blockages within the vasculature, by addressing the frictional resistance on the obstruction prior to attempting to translate and/or mobilize the obstruction within the body lumen.

22 Claims, 27 Drawing Sheets



Related U.S. Application Data

Jun. 6, 2016, now Pat. No. 9,943,323, which is a continuation of application No. 14/446,755, filed on Jul. 30, 2014, now Pat. No. 9,358,094, which is a continuation of application No. 13/959,433, filed on Aug. 5, 2013, now Pat. No. 8,795,305, which is a continuation of application No. PCT/US2012/039216, filed on May 23, 2012.

(60) Provisional application No. 61/489,254, filed on May 24, 2011, provisional application No. 61/489,183, filed on May 23, 2011.

(51) **Int. Cl.**
A61F 2/962 (2013.01)
A61B 17/34 (2006.01)

(56) **References Cited**

U.S. PATENT DOCUMENTS

3,996,938	A	12/1976	Clark
4,347,846	A	9/1982	Dormia
4,611,594	A	9/1986	Grayhack et al.
4,650,466	A	3/1987	Luther
4,657,020	A	4/1987	Lifton
4,699,147	A	10/1987	Chilson et al.
4,790,812	A	12/1988	Hawkins et al.
4,807,626	A	2/1989	McGirr
4,832,055	A	5/1989	Palestrant
4,873,978	A	10/1989	Ginsburg
4,969,891	A	11/1990	Gewertz
4,998,539	A	3/1991	Delsanti
5,034,001	A	7/1991	Garrison et al.
5,057,114	A	10/1991	Wittich et al.
5,059,178	A	10/1991	Ya
5,102,415	A	4/1992	Guenther et al.
5,147,400	A	9/1992	Kaplan et al.
5,152,777	A	10/1992	Goldberg et al.
5,192,286	A	3/1993	Phan et al.
5,300,086	A	4/1994	Gory et al.
5,329,942	A	7/1994	Gunther et al.
5,443,478	A	8/1995	Purdy
5,449,372	A	9/1995	Schmaltz et al.
5,458,375	A	10/1995	Anspach et al.
5,490,859	A	2/1996	Mische et al.
5,496,330	A	3/1996	Bates et al.
5,509,900	A	4/1996	Kirkman
5,653,684	A	8/1997	Laptewicz et al.
5,658,296	A	8/1997	Bates et al.
5,709,704	A	1/1998	Nott et al.
5,733,302	A	3/1998	Myler et al.
5,741,325	A	4/1998	Chaikof et al.
5,792,156	A	8/1998	Perouse
5,827,324	A	10/1998	Cassell et al.
5,846,251	A	12/1998	Hart
5,895,398	A	4/1999	Wensel et al.
5,941,869	A	8/1999	Patterson et al.
5,947,995	A	9/1999	Samuels
5,968,090	A	10/1999	Ratcliff et al.
5,971,938	A	10/1999	Hart et al.
5,972,019	A	10/1999	Engelson et al.
5,984,957	A	11/1999	Laptewicz et al.
6,001,118	A	12/1999	Daniel et al.
6,033,394	A	3/2000	Vidlund et al.
6,042,598	A	3/2000	Tsugita et al.
6,053,932	A	4/2000	Daniel et al.
6,066,149	A	5/2000	Samson et al.
6,066,158	A	5/2000	Engelson et al.
6,096,053	A	8/2000	Bates
6,099,534	A	8/2000	Bates et al.
6,146,403	A	11/2000	St
6,159,220	A	12/2000	Gobron et al.
6,165,200	A	12/2000	Tsugita et al.
6,168,603	B1	1/2001	Leslie et al.
6,174,318	B1	1/2001	Bates et al.
6,176,873	B1	1/2001	Ouchi
6,190,394	B1	2/2001	Lind et al.
6,217,609	B1	4/2001	Haverkost
6,221,006	B1	4/2001	Dubrul et al.
6,245,088	B1	6/2001	Lowery
6,245,089	B1	6/2001	Daniel et al.
6,248,113	B1	6/2001	Fina
6,264,664	B1	7/2001	Avellanet
6,302,895	B1	10/2001	Gobron et al.
6,309,399	B1	10/2001	Barbut et al.
6,348,056	B1	2/2002	Bates et al.
6,350,266	B1	2/2002	White et al.
6,364,895	B1	4/2002	Greenhalgh
6,371,971	B1	4/2002	Tsugita et al.
6,383,195	B1	5/2002	Richard
6,383,196	B1	5/2002	Leslie et al.
6,391,044	B1	5/2002	Yadav et al.
6,402,771	B1	6/2002	Palmer et al.
6,409,750	B1	6/2002	Hyodoh et al.
6,416,505	B1	7/2002	Fleischman et al.
6,425,909	B1	7/2002	Dieck et al.
6,436,112	B2	8/2002	Wensel et al.
6,443,972	B1	9/2002	Bosma et al.
6,458,139	B1	10/2002	Palmer et al.
6,485,497	B2	11/2002	Wensel et al.
6,494,884	B2	12/2002	Gifford et al.
6,506,204	B2	1/2003	Mazzocchi
6,514,273	B1	2/2003	Voss et al.
6,530,935	B2	3/2003	Wensel et al.
6,540,657	B2	4/2003	Cross et al.
6,540,768	B1	4/2003	Diaz et al.
6,551,342	B1	4/2003	Shen et al.
6,575,997	B1	6/2003	Palmer et al.
6,585,753	B2	7/2003	Eder et al.
6,592,605	B2	7/2003	Lenker et al.
6,592,607	B1	7/2003	Palmer et al.
6,602,271	B2	8/2003	Adams et al.
6,605,102	B1	8/2003	Mazzocchi et al.
6,610,077	B1	8/2003	Hancock et al.
6,616,679	B1	9/2003	Khosravi et al.
6,620,148	B1	9/2003	Tsugita
6,635,068	B1	10/2003	Dubrul et al.
6,636,758	B2	10/2003	Sanchez et al.
6,638,245	B2	10/2003	Miller et al.
6,638,293	B1	10/2003	Makower et al.
6,641,590	B1	11/2003	Palmer et al.
6,645,199	B1	11/2003	Jenkins et al.
6,652,505	B1	11/2003	Tsugita
6,652,548	B2	11/2003	Evans et al.
6,660,021	B1	12/2003	Palmer et al.
6,663,650	B2	12/2003	Sepetka et al.
6,673,042	B1	1/2004	Samson et al.
6,679,893	B1	1/2004	Tran
6,685,738	B2	2/2004	Chouinard et al.
6,692,508	B2	2/2004	Wensel et al.
6,692,509	B2	2/2004	Wensel et al.
6,695,858	B1	2/2004	Dubrul et al.
6,702,782	B2	3/2004	Miller et al.
6,730,104	B1	5/2004	Sepetka et al.
6,745,080	B2	6/2004	Koblish
6,746,468	B1	6/2004	Sepetka et al.
6,749,619	B2	6/2004	Ouriel et al.
6,755,813	B2	6/2004	Ouriel et al.
6,800,080	B1	10/2004	Bates
6,824,545	B2	11/2004	Sepetka et al.
6,855,155	B2	2/2005	Denardo et al.
6,872,211	B2	3/2005	White et al.
6,872,216	B2	3/2005	Daniel et al.
6,890,341	B2	5/2005	Dieck et al.
6,893,431	B2	5/2005	Naimark et al.
6,905,503	B2	6/2005	Gifford et al.
6,913,612	B2	7/2005	Palmer et al.
6,936,059	B2	8/2005	Belef
6,939,362	B2	9/2005	Boyle et al.
6,945,977	B2	9/2005	Demarais et al.
6,953,465	B2	10/2005	Dieck et al.
6,964,672	B2	11/2005	Brady et al.
7,004,955	B2	2/2006	Shen et al.
7,004,956	B2	2/2006	Palmer et al.

(56) **References Cited**

U.S. PATENT DOCUMENTS

7,037,320	B2	5/2006	Brady et al.
7,041,126	B2	5/2006	Shin et al.
7,048,014	B2	5/2006	Hyodoh et al.
7,058,456	B2	6/2006	Pierce
7,097,653	B2	8/2006	Freudenthal et al.
7,101,380	B2	9/2006	Khachin et al.
7,169,165	B2	1/2007	Belef et al.
7,179,273	B1	2/2007	Palmer et al.
7,182,771	B1	2/2007	Houser et al.
7,235,061	B2	6/2007	Tsugita
7,240,516	B2	7/2007	Pryor
7,399,308	B2	7/2008	Borillo et al.
7,534,252	B2	5/2009	Sepetka et al.
7,578,830	B2	8/2009	Kusleika et al.
7,621,870	B2	11/2009	Berrada et al.
7,837,702	B2	11/2010	Bates
8,070,791	B2	12/2011	Ferrera et al.
8,088,140	B2	1/2012	Ferrera et al.
8,105,333	B2	1/2012	Sepetka et al.
8,197,493	B2	6/2012	Ferrera et al.
8,603,014	B2	12/2013	Alleman et al.
8,795,305	B2	8/2014	Martin et al.
8,837,800	B1	9/2014	Bammer et al.
8,932,319	B2	1/2015	Martin et al.
9,119,656	B2	9/2015	Bose et al.
9,126,018	B1	9/2015	Garrison
9,211,132	B2	12/2015	Bowman
9,241,699	B1	1/2016	Kume et al.
9,265,512	B2	2/2016	Garrison et al.
9,308,007	B2	4/2016	Cully et al.
9,358,094	B2	6/2016	Martin et al.
9,399,118	B2	7/2016	Kume et al.
9,445,828	B2	9/2016	Turjman et al.
9,445,829	B2	9/2016	Brady et al.
9,492,637	B2	11/2016	Garrison et al.
9,539,022	B2	1/2017	Bowman
9,561,345	B2	2/2017	Garrison et al.
9,579,119	B2	2/2017	Cully et al.
9,585,741	B2	3/2017	Ma
9,642,635	B2	5/2017	Vale et al.
9,655,633	B2	5/2017	Leynov et al.
9,737,318	B2	8/2017	Monstadt et al.
9,770,251	B2	9/2017	Bowman et al.
9,801,643	B2	10/2017	Hansen et al.
9,861,783	B2	1/2018	Garrison et al.
9,943,323	B2	4/2018	Martin et al.
9,993,257	B2	6/2018	Losordo et al.
10,028,782	B2	7/2018	Orion
10,029,008	B2	7/2018	Creighton
10,039,906	B2	8/2018	Kume et al.
11,529,155	B2	12/2022	Martin et al.
2001/0041909	A1	11/2001	Tsugita et al.
2001/0044632	A1	11/2001	Daniel et al.
2001/0044634	A1	11/2001	Don et al.
2001/0051810	A1	12/2001	Dubrul et al.
2002/0002396	A1	1/2002	Fulkerson
2002/0004667	A1	1/2002	Adams et al.
2002/0026211	A1	2/2002	Khosravi et al.
2002/0058904	A1	5/2002	Boock et al.
2002/0062135	A1	5/2002	Mazzocchi et al.
2002/0072764	A1	6/2002	Sepetka et al.
2002/0082558	A1	6/2002	Samson et al.
2002/0123765	A1	9/2002	Sepetka et al.
2002/0138094	A1	9/2002	Borillo et al.
2002/0151928	A1	10/2002	Leslie et al.
2002/0169474	A1	11/2002	Kusleika et al.
2002/0188314	A1	12/2002	Anderson et al.
2002/0193825	A1	12/2002	McGuckin et al.
2003/0004542	A1	1/2003	Wensel et al.
2003/0023265	A1	1/2003	Forber
2003/0040771	A1	2/2003	Hyodoh et al.
2003/0050663	A1	3/2003	Khachin et al.
2003/0060782	A1	3/2003	Bose et al.
2003/0093087	A1	5/2003	Jones et al.
2003/0144687	A1	7/2003	Brady et al.
2003/0153935	A1	8/2003	Mialhe
2003/0176884	A1*	9/2003	Berrada A61F 2/0105 606/200
2003/0187495	A1*	10/2003	Cully A61B 17/12109 623/1.34
2003/0195556	A1	10/2003	Stack et al.
2003/0199921	A1	10/2003	Palmer et al.
2004/0068288	A1	4/2004	Palmer et al.
2004/0073243	A1	4/2004	Sepetka et al.
2004/0079429	A1	4/2004	Miller et al.
2004/0133232	A1	7/2004	Rosenbluth et al.
2004/0138692	A1	7/2004	Phung et al.
2004/0153025	A1	8/2004	Seifert et al.
2004/0153118	A1	8/2004	Clubb et al.
2004/0172056	A1	9/2004	Guterman et al.
2004/0199201	A1	10/2004	Kellett et al.
2004/0199243	A1	10/2004	Yodfat
2004/0210116	A1	10/2004	Nakao
2004/0267301	A1	12/2004	Boylan et al.
2005/0004594	A1	1/2005	Nool et al.
2005/0033348	A1	2/2005	Sepetka et al.
2005/0038447	A1	2/2005	Huffmaster
2005/0043680	A1	2/2005	Segal et al.
2005/0043756	A1	2/2005	Lavelle et al.
2005/0049619	A1	3/2005	Sepetka et al.
2005/0055033	A1	3/2005	Leslie et al.
2005/0055047	A1	3/2005	Greenhalgh
2005/0059995	A1	3/2005	Sepetka et al.
2005/0080356	A1	4/2005	Dapolito et al.
2005/0085826	A1	4/2005	Nair et al.
2005/0085847	A1	4/2005	Galdonik et al.
2005/0085849	A1	4/2005	Sepetka et al.
2005/0090857	A1	4/2005	Kusleika et al.
2005/0090858	A1	4/2005	Pavlovic
2005/0125024	A1	6/2005	Sepetka et al.
2005/0131450	A1	6/2005	Nicholson et al.
2005/0171566	A1	8/2005	Kanamaru
2005/0203571	A1	9/2005	Mazzocchi et al.
2005/0209609	A1	9/2005	Wallace
2005/0216030	A1	9/2005	Sepetka et al.
2005/0216050	A1	9/2005	Sepetka et al.
2005/0234501	A1	10/2005	Barone
2005/0234505	A1	10/2005	Diaz et al.
2005/0277978	A1	12/2005	Greenhalgh
2005/0283166	A1	12/2005	Greenhalgh
2005/0283186	A1	12/2005	Berrada et al.
2006/0004404	A1	1/2006	Khachin et al.
2006/0009784	A1	1/2006	Behl et al.
2006/0030925	A1	2/2006	Pryor
2006/0047286	A1	3/2006	West
2006/0058836	A1	3/2006	Bose et al.
2006/0058837	A1	3/2006	Bose et al.
2006/0058838	A1	3/2006	Bose et al.
2006/0095070	A1	5/2006	Gilson et al.
2006/0129166	A1	6/2006	Lavelle
2006/0129180	A1	6/2006	Tsugita et al.
2006/0155305	A1	7/2006	Freudenthal et al.
2006/0190070	A1	8/2006	Dieck et al.
2006/0195137	A1	8/2006	Sepetka et al.
2006/0229638	A1	10/2006	Abrams et al.
2006/0253145	A1	11/2006	Lucas
2006/0271153	A1	11/2006	Garcia et al.
2006/0276805	A1	12/2006	Yu
2006/0282111	A1	12/2006	Morsi
2006/0287668	A1*	12/2006	Fawzi A61F 2/0105 606/200
2007/0112374	A1	5/2007	Paul et al.
2007/0118165	A1	5/2007	Demello et al.
2007/0149996	A1	6/2007	Coughlin
2007/0185500	A1	8/2007	Martin et al.
2007/0185501	A1	8/2007	Martin et al.
2007/0197103	A1	8/2007	Martin et al.
2007/0198029	A1	8/2007	Martin et al.
2007/0198030	A1	8/2007	Martin et al.
2007/0198051	A1	8/2007	Clubb et al.
2007/0225749	A1	9/2007	Martin et al.
2007/0233236	A1	10/2007	Pryor
2007/0265656	A1	11/2007	Amplatz et al.

(56) **References Cited**

U.S. PATENT DOCUMENTS

2007/0299465	A1*	12/2007	Messal	A61F 2/011 606/200
2008/0103522	A1*	5/2008	Steingisser	A61F 2/01 606/200
2008/0109031	A1	5/2008	Sepetka et al.	
2008/0183198	A1	7/2008	Sepetka et al.	
2008/0188885	A1	8/2008	Sepetka et al.	
2008/0262528	A1	10/2008	Martin	
2008/0262532	A1	10/2008	Martin	
2009/0069828	A1	3/2009	Martin et al.	
2009/0105722	A1	4/2009	Fulkerson et al.	
2009/0105737	A1	4/2009	Fulkerson et al.	
2009/0125053	A1	5/2009	Ferrera et al.	
2009/0192518	A1	7/2009	Leanna et al.	
2009/0287291	A1	11/2009	Becking et al.	
2009/0299393	A1	12/2009	Martin et al.	
2010/0076452	A1	3/2010	Sepetka et al.	
2010/0100106	A1	4/2010	Ferrera	
2010/0174309	A1	7/2010	Fulkerson et al.	
2010/0185210	A1	7/2010	Hauser et al.	
2010/0217187	A1	8/2010	Ferrera et al.	
2010/0256600	A1	10/2010	Ferrera	
2010/0318097	A1	12/2010	Cragg et al.	
2011/0060359	A1	3/2011	Hannes et al.	
2011/0152920	A1	6/2011	Eckhouse et al.	
2011/0160742	A1	6/2011	Ferrera et al.	
2011/0160757	A1	6/2011	Ferrera et al.	
2011/0160760	A1	6/2011	Ferrera et al.	
2011/0160761	A1	6/2011	Ferrera et al.	
2011/0160763	A1	6/2011	Ferrera et al.	
2011/0166586	A1	7/2011	Sepetka et al.	
2011/0288572	A1	11/2011	Martin	
2011/0319917	A1	12/2011	Ferrera et al.	
2012/0143230	A1	6/2012	Sepetka et al.	
2012/0197285	A1	8/2012	Martin et al.	
2013/0030461	A1	1/2013	Marks et al.	
2013/0281788	A1	10/2013	Garrison	
2014/0276074	A1	9/2014	Warner	
2014/0343595	A1	11/2014	Monstadt et al.	
2015/0359547	A1	12/2015	Vale et al.	
2016/0015402	A1	1/2016	Brady et al.	
2016/0015935	A1	1/2016	Chan et al.	
2016/0106448	A1	4/2016	Brady et al.	
2016/0106449	A1	4/2016	Brady et al.	
2016/0113663	A1	4/2016	Brady et al.	
2016/0113665	A1	4/2016	Brady et al.	
2016/0151618	A1	6/2016	Powers et al.	
2016/0157985	A1	6/2016	Vo et al.	
2016/0199620	A1	7/2016	Pokorney et al.	
2016/0296690	A1	10/2016	Kume et al.	
2016/0302808	A1	10/2016	Loganathan et al.	
2016/0354098	A1	12/2016	Martin et al.	
2016/0375180	A1	12/2016	Anzai	
2017/0079766	A1	3/2017	Wang et al.	
2017/0079767	A1	3/2017	Leon-Yip	
2017/0086862	A1	3/2017	Vale et al.	
2017/0100143	A1	4/2017	Grandfield	
2017/0105743	A1	4/2017	Vale et al.	
2017/0164963	A1	6/2017	Goyal	
2017/0215902	A1	8/2017	Leynov et al.	
2017/0224953	A1	8/2017	Tran et al.	
2017/0281909	A1	10/2017	Northrop et al.	
2017/0290599	A1	10/2017	Youn et al.	
2018/0049762	A1	2/2018	Seip et al.	
2018/0084982	A1	3/2018	Yamashita et al.	
2018/0116717	A1	5/2018	Taff et al.	
2018/0132876	A1	5/2018	Zaidat	
2018/0140314	A1	5/2018	Goyal et al.	
2018/0140315	A1	5/2018	Bowman et al.	
2018/0140354	A1	5/2018	Lam et al.	
2018/0185614	A1	7/2018	Garrison et al.	
2019/0374239	A1	12/2019	Martin et al.	

FOREIGN PATENT DOCUMENTS

DE	3501707 A1		7/1986	
DE	60038501	*	4/2009	
EP	0200668 A2		11/1986	
EP	1312314 A1		5/2003	
EP	2319575 B1		11/2013	
JP	2002537943 A		11/2002	
JP	2007522881 A		8/2007	
JP	2007252951 A		10/2007	
JP	2008539958 A		11/2008	
JP	2011508635 A		3/2011	
JP	2014004219 A		1/2014	
JP	2018118132 A		8/2018	
KR	20180102877 A		9/2018	
WO	9409845 A1		5/1994	
WO	9509586 A1		4/1995	
WO	9601591 A1		1/1996	
WO	9617634 A2		6/1996	
WO	9619941 A1		7/1996	
WO	9727808 A1		8/1997	
WO	9727893 A1		8/1997	
WO	9803120 A1		1/1998	
WO	0053120 A1		9/2000	
WO	0072909 A1		12/2000	
WO	0132254 A1		5/2001	
WO	0154622 A1		8/2001	
WO	0167967 A1		9/2001	
WO	0202162 A2		1/2002	
WO	0228291 A2		4/2002	
WO	03000334 A1		1/2003	
WO	03061730 A2		7/2003	
WO	03089039 A1		10/2003	
WO	WO 2005027751	*	3/2005	
WO	WO-2005027751 A1	*	3/2005	A61B 17/221
WO	2006031410 A2		3/2006	
WO	2006122076 A1		11/2006	
WO	2007092820 A2		8/2007	
WO	2008036156 A1		3/2008	
WO	2008131116 A1		10/2008	
WO	2009034456 A2		3/2009	
WO	2009086482 A1		7/2009	
WO	2011091383 A1		7/2011	
WO	2012009675 A2		1/2012	
WO	2012162437 A1		11/2012	
WO	2013106146 A1		7/2013	
WO	2015141317 A1		9/2015	
WO	2017192999 A1		11/2017	
WO	2018019829 A1		2/2018	
WO	2018033401 A1		2/2018	
WO	2018046408 A2		3/2018	
WO	2018137029 A1		8/2018	
WO	2018137030 A1		8/2018	
WO	2018145212 A1		8/2018	
WO	2018156813 A1		8/2018	
WO	2018172891 A1		9/2018	
WO	2018187776 A1		10/2018	

* cited by examiner

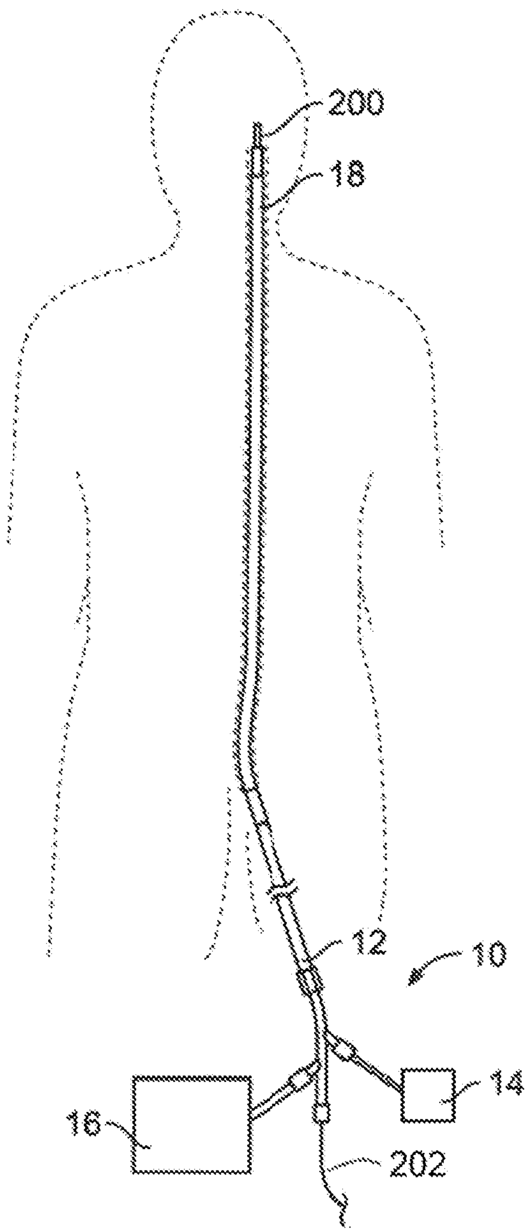
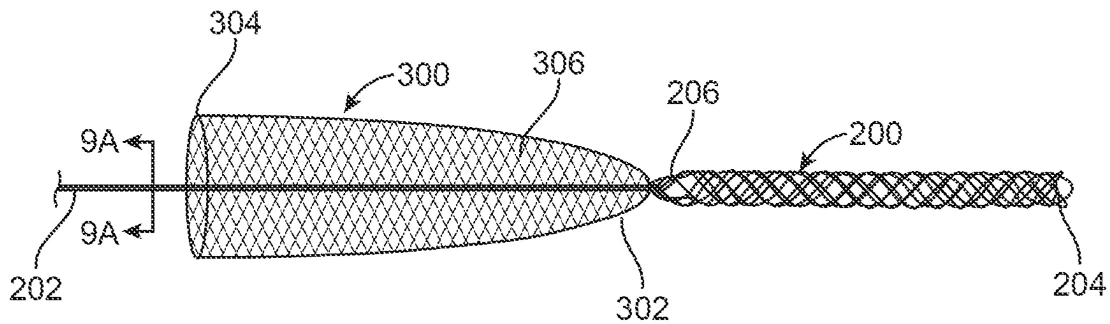


FIG.1



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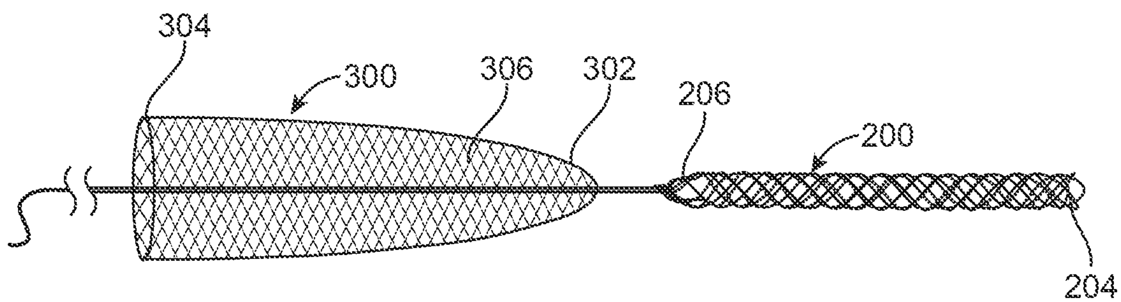

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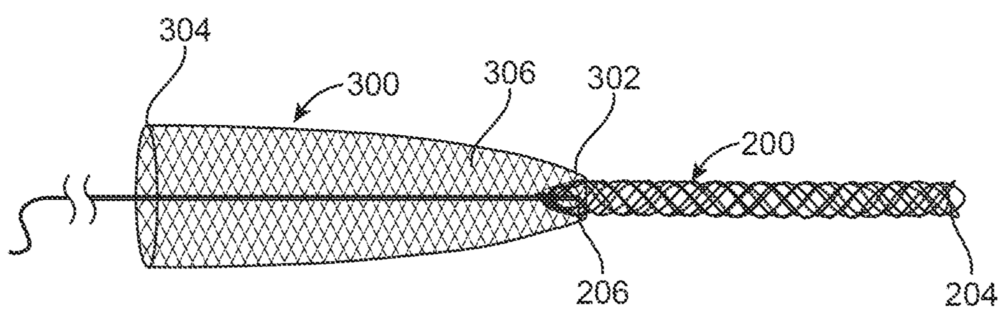


FIG. 2C

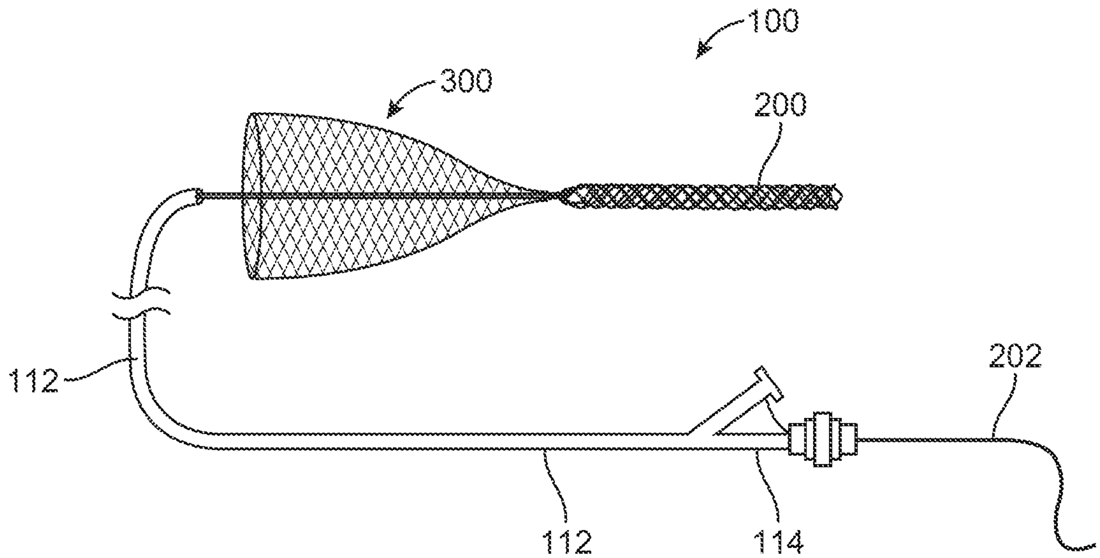


FIG. 2D

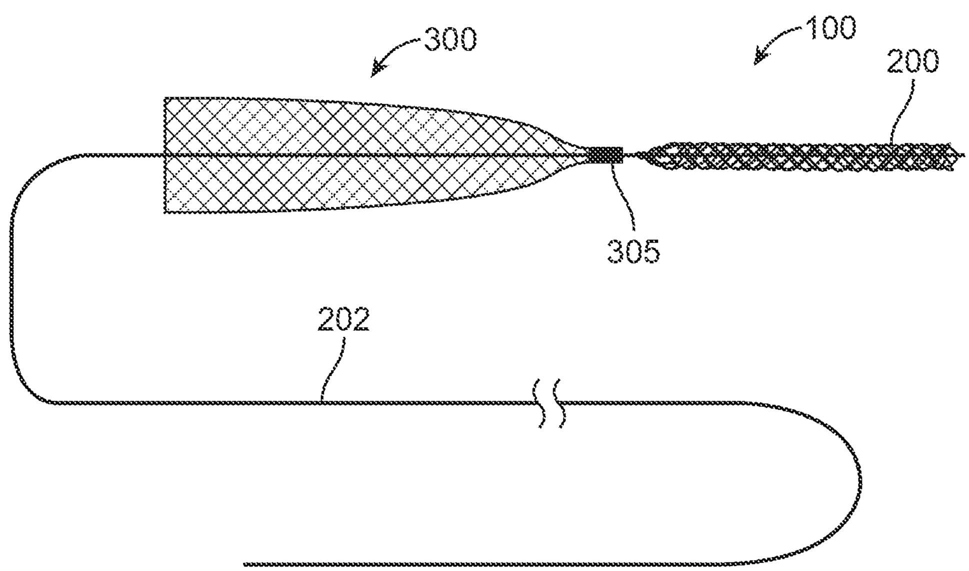


FIG. 2E

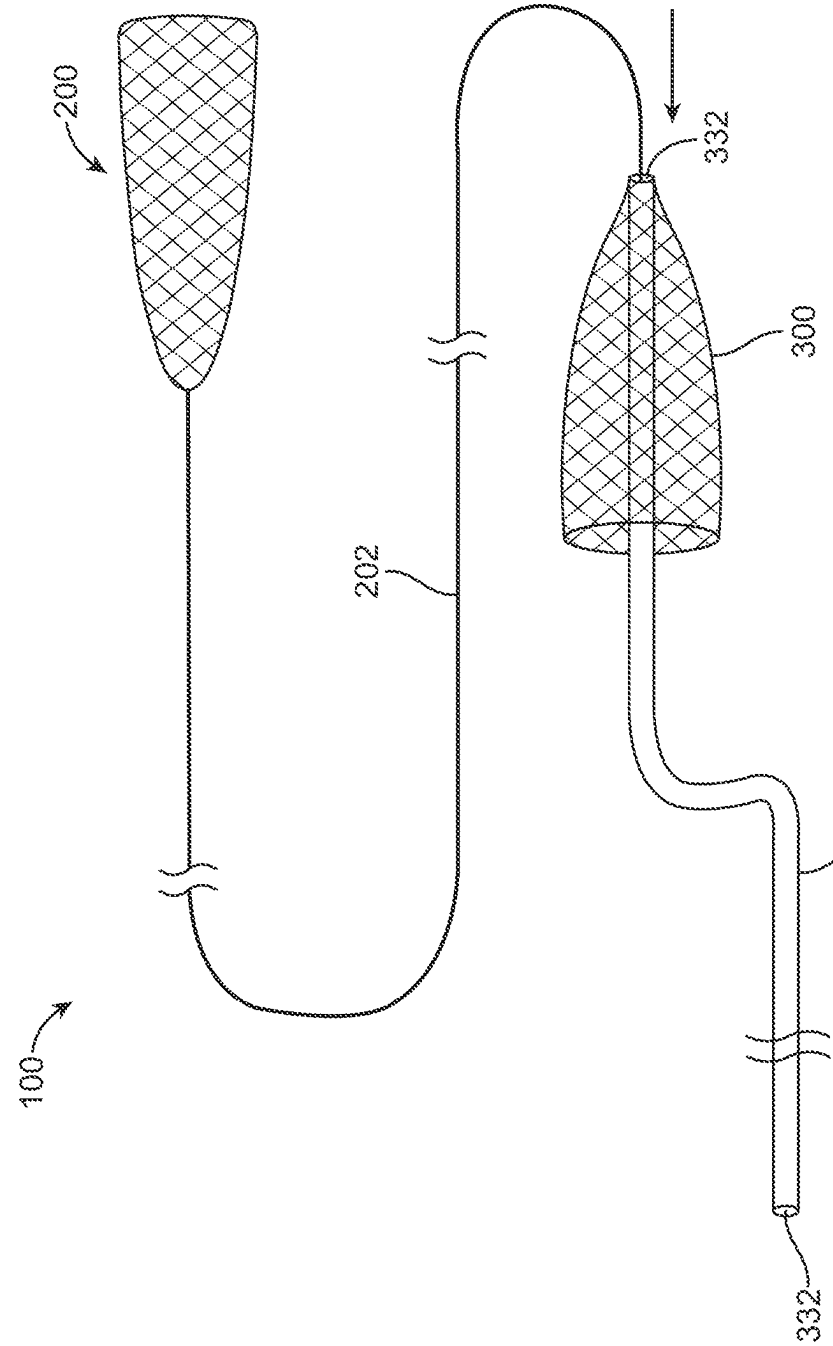


FIG. 2F

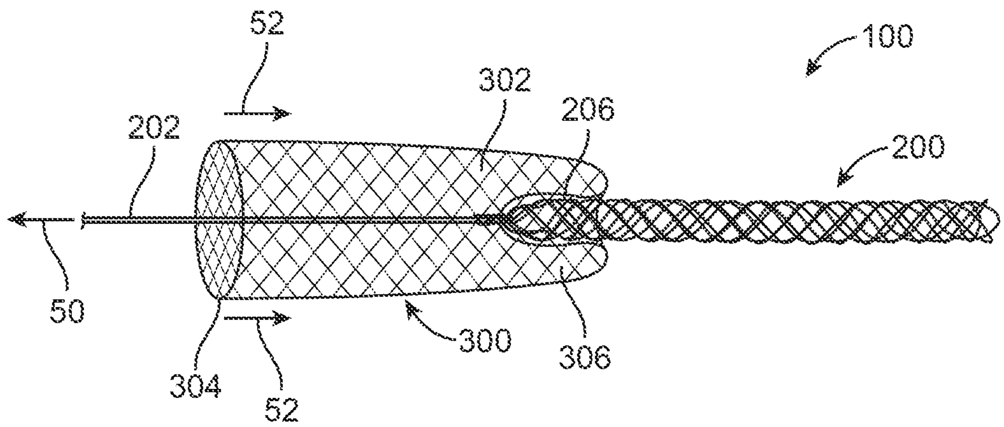


FIG. 3A

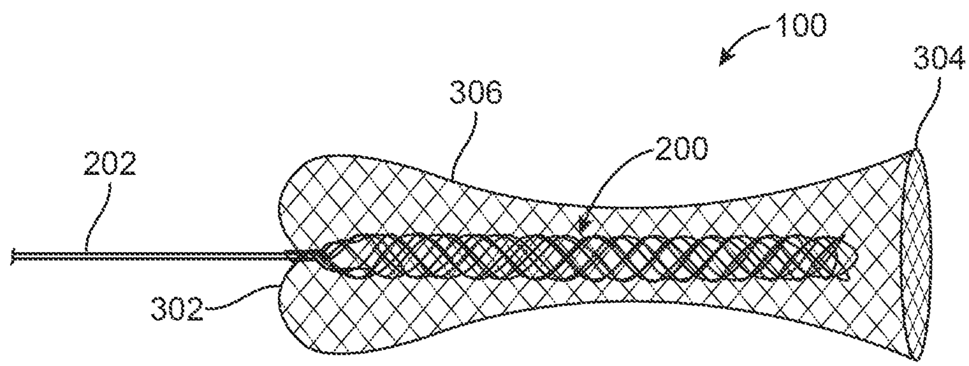


FIG. 3B

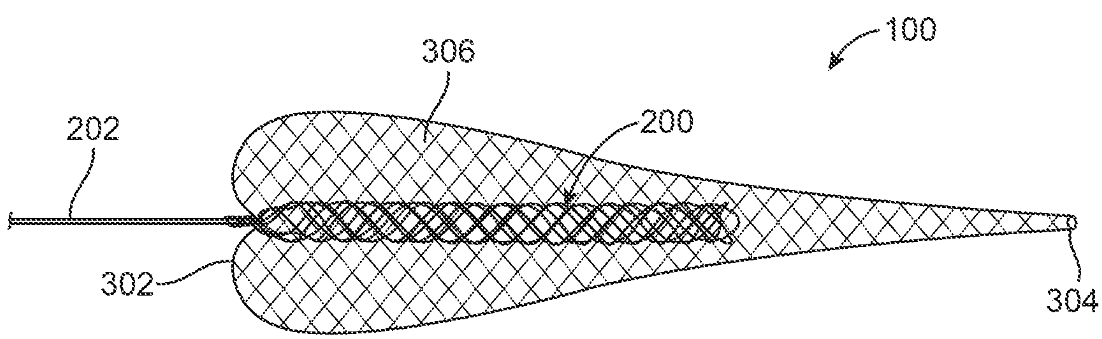


FIG. 3C

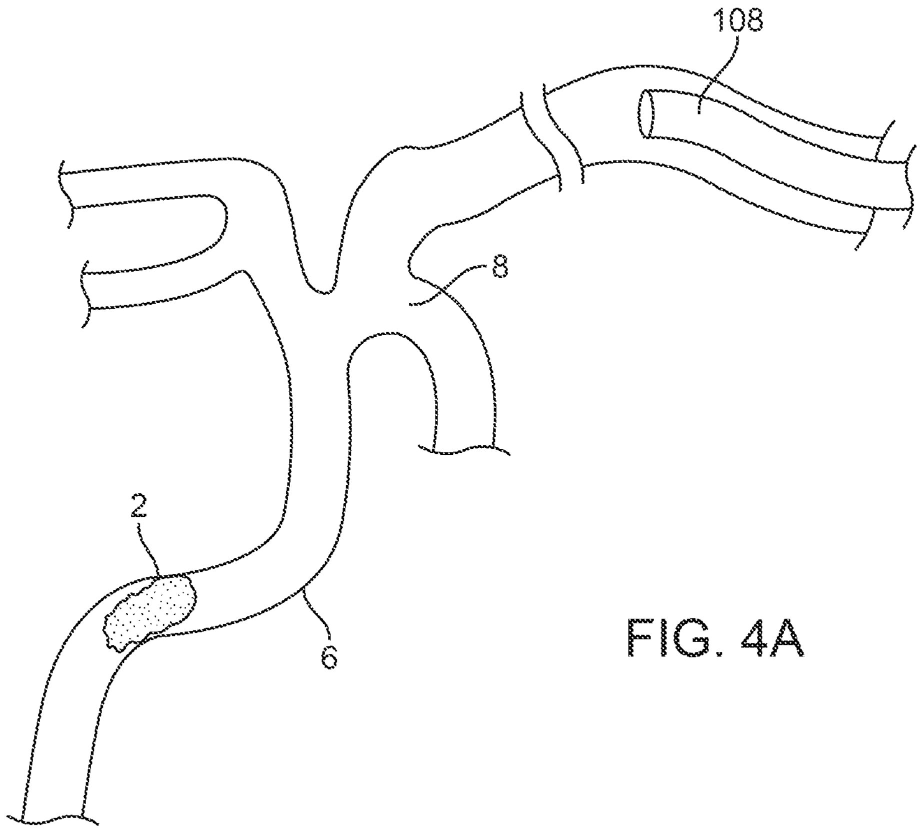
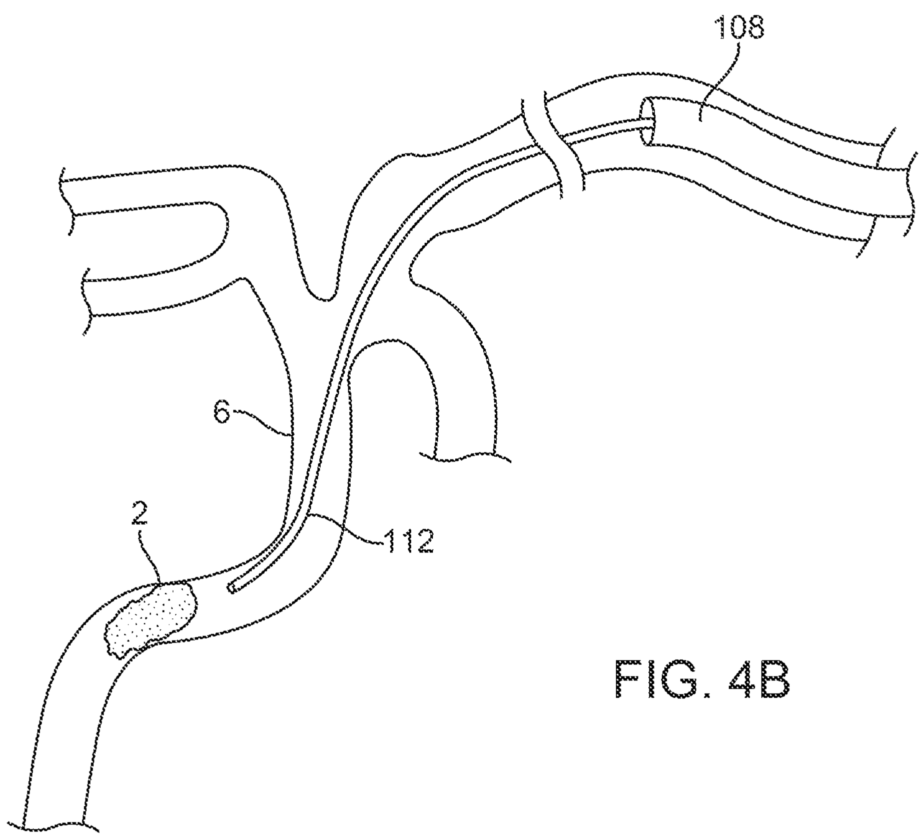


FIG. 4A



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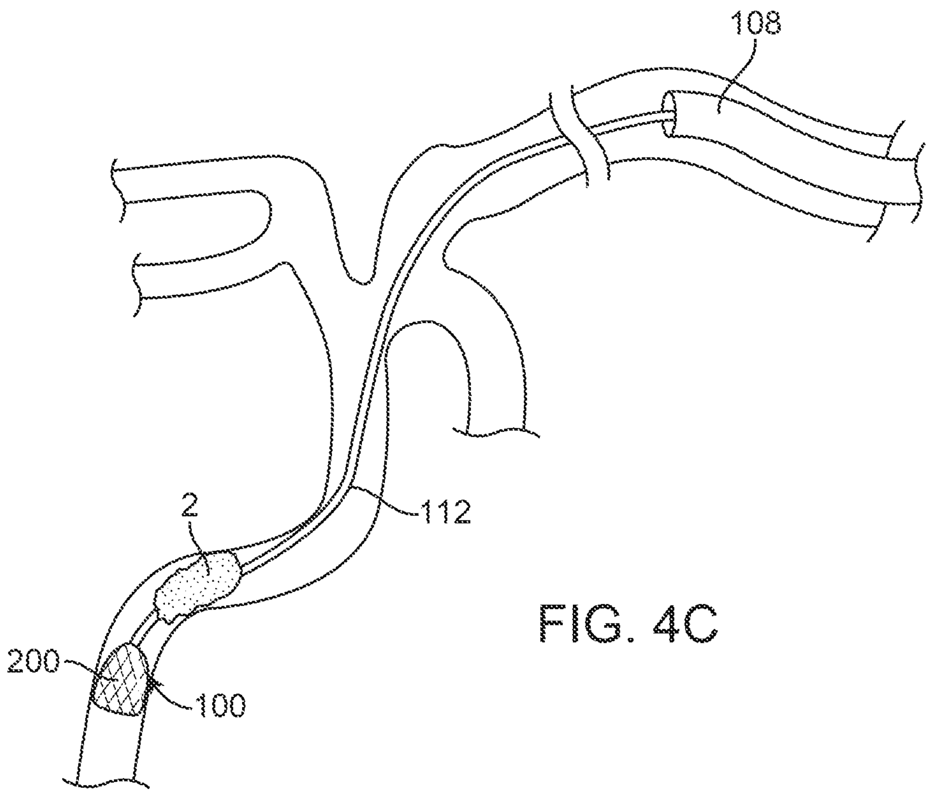


FIG. 4C

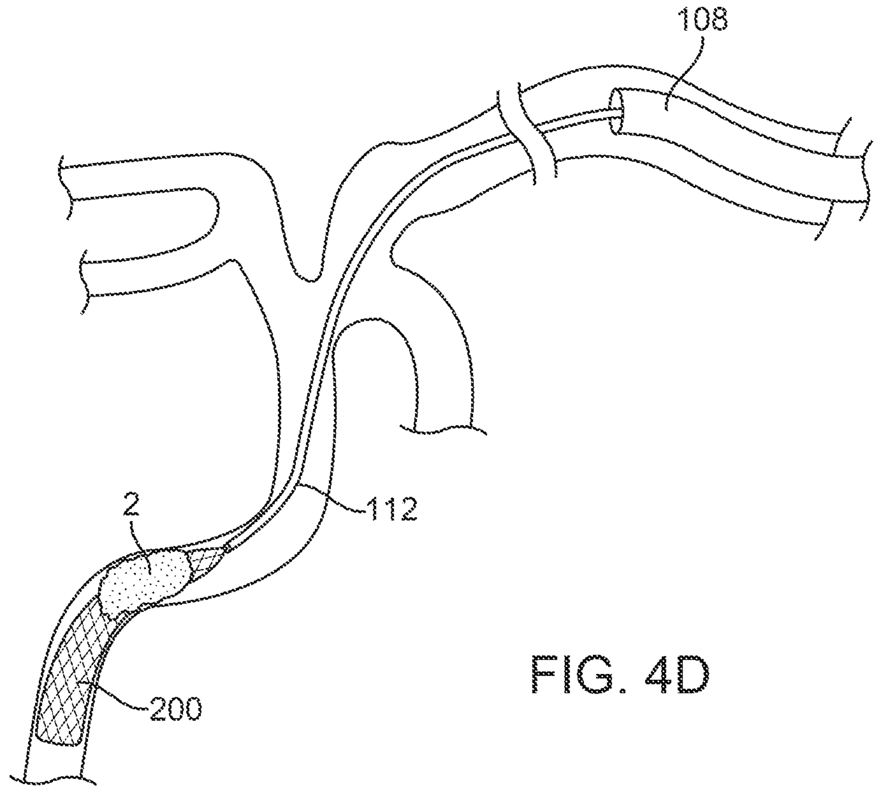


FIG. 4D

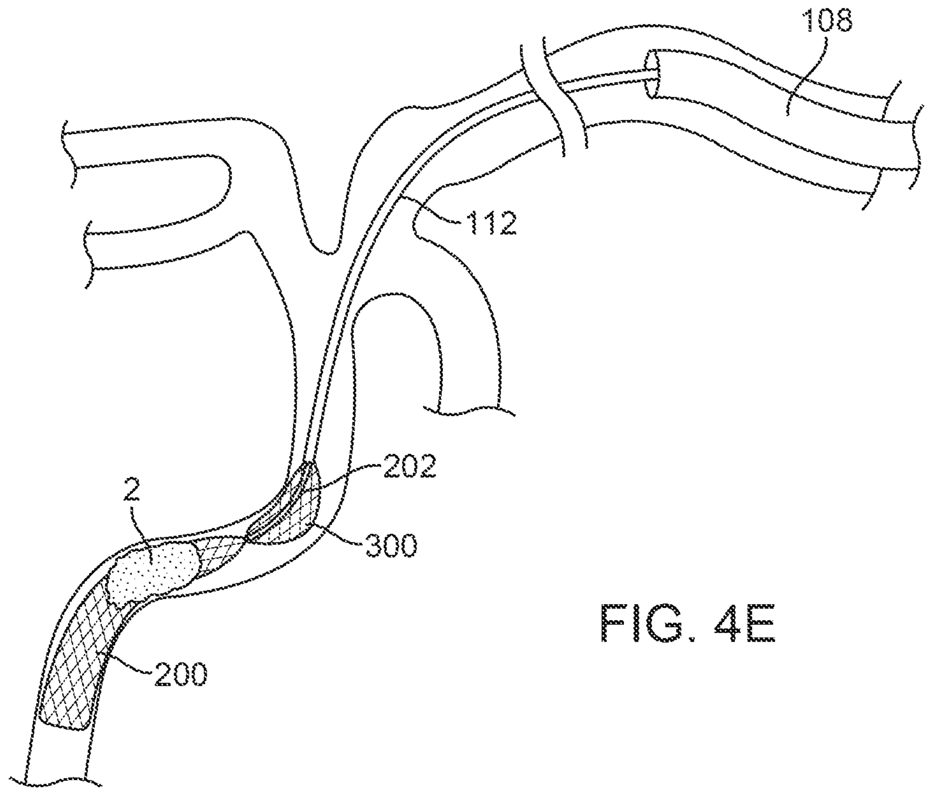


FIG. 4E

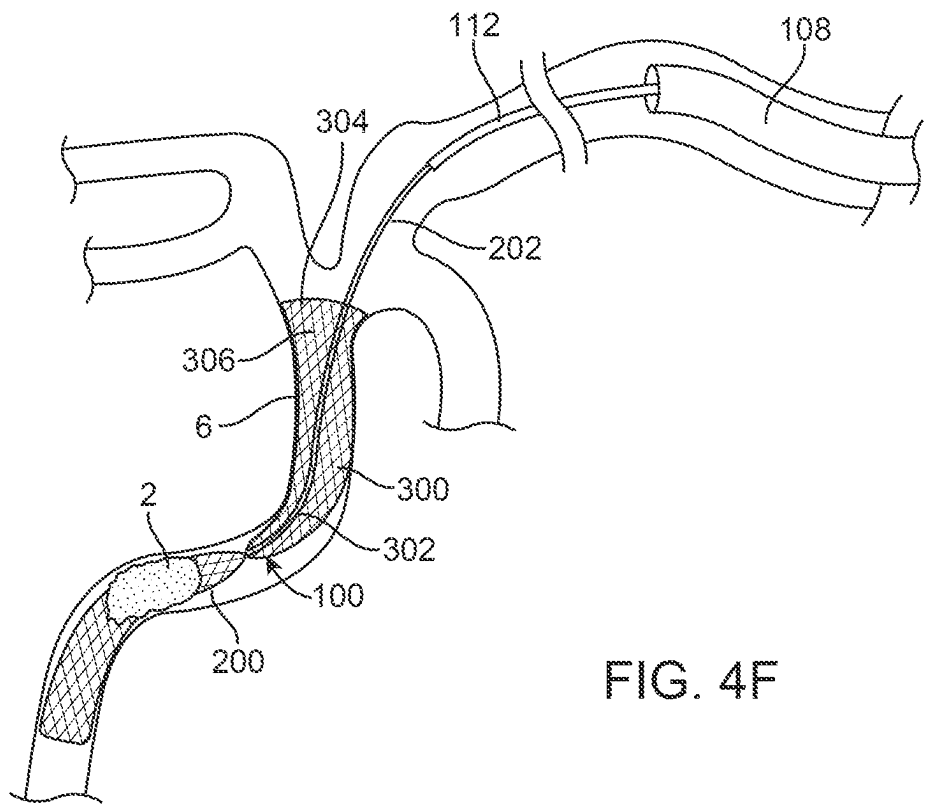


FIG. 4F

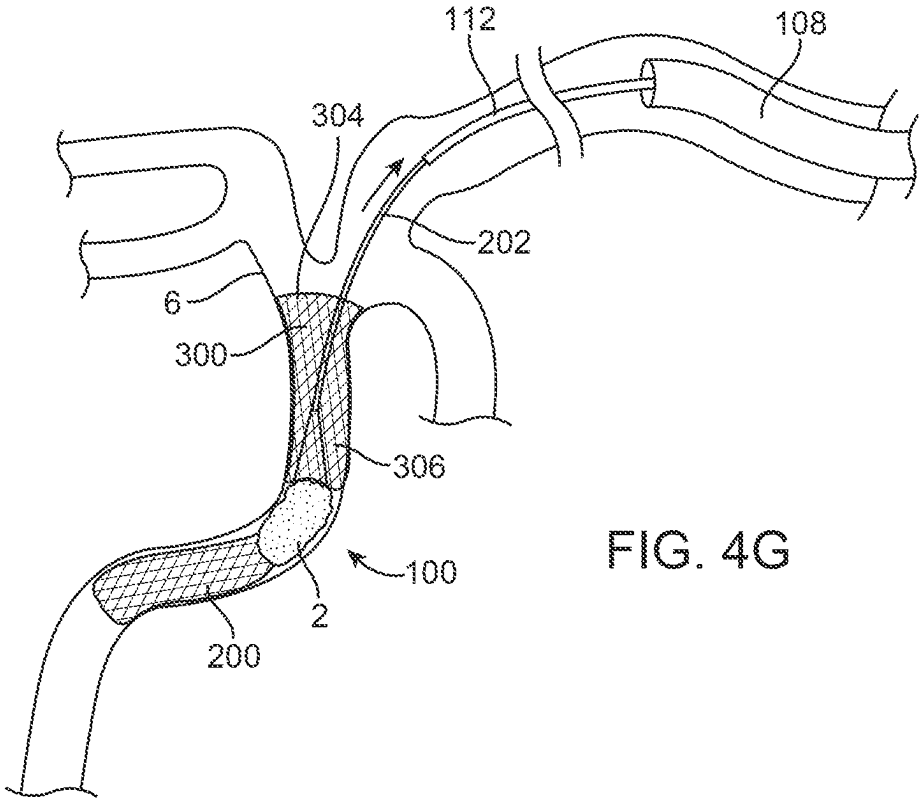


FIG. 4G

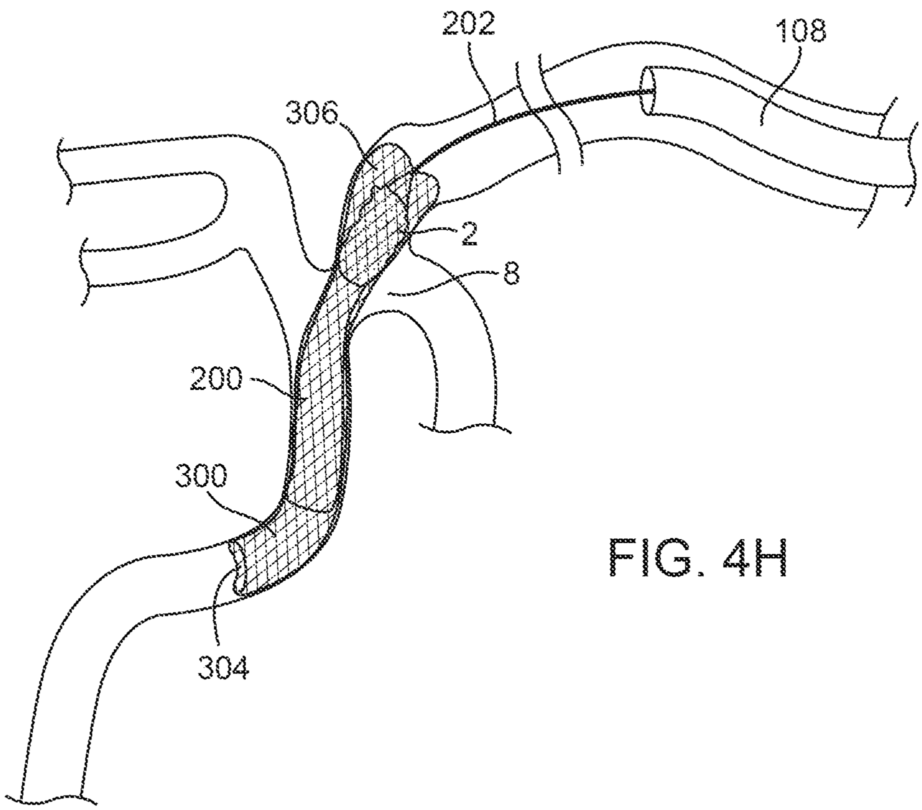


FIG. 4H

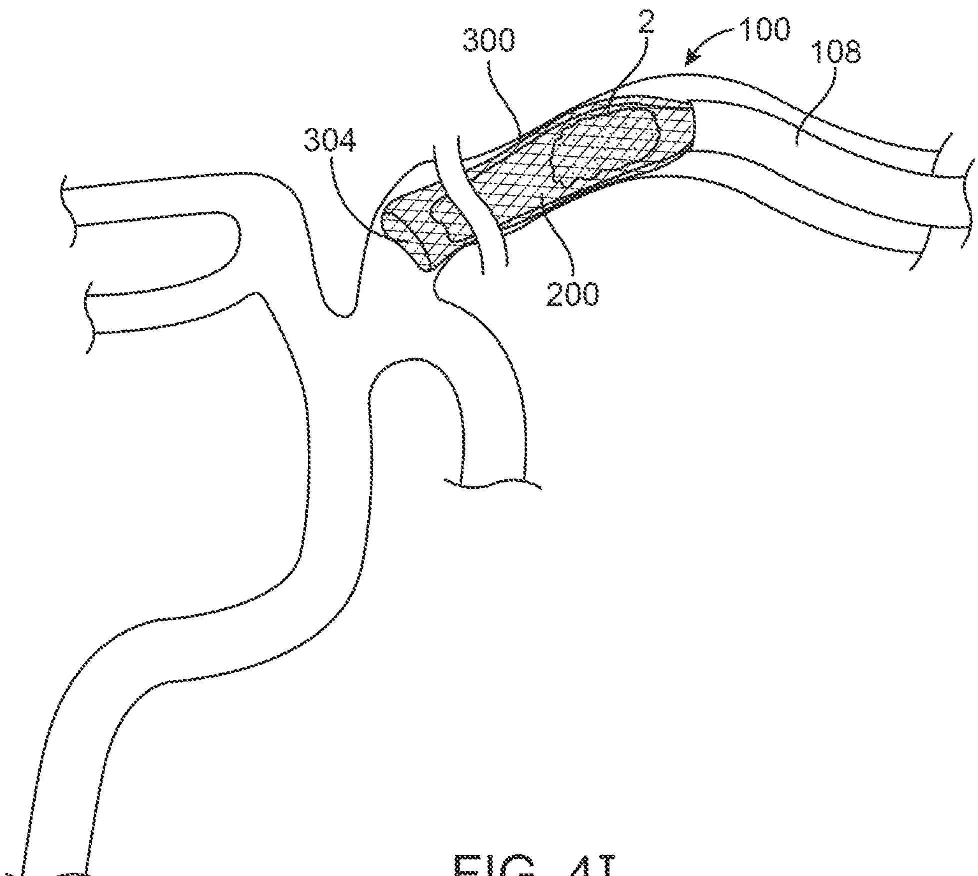


FIG. 4I

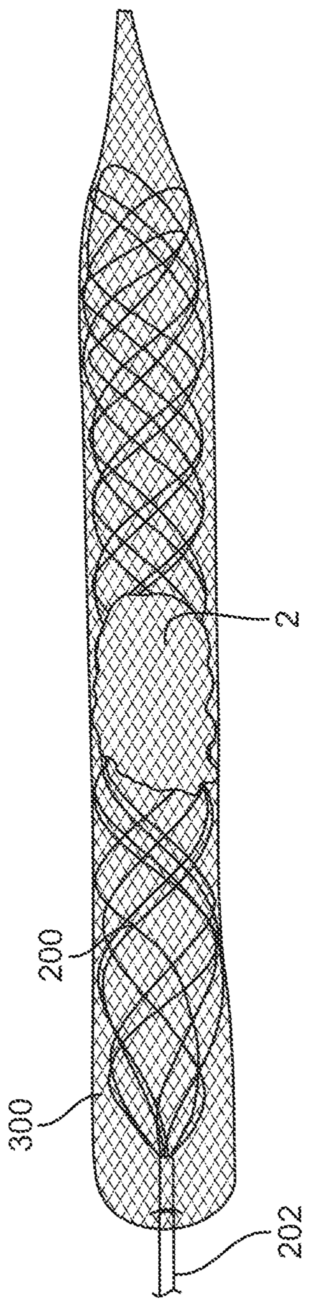


FIG. 4J

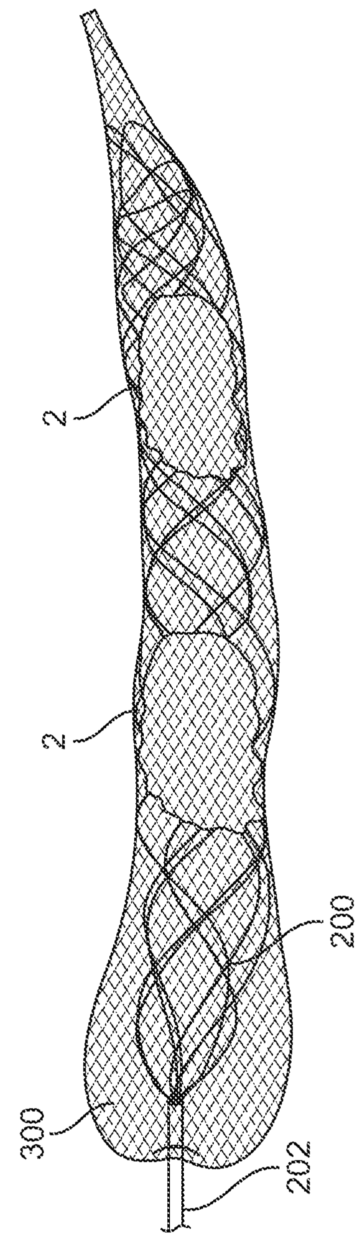


FIG. 4K

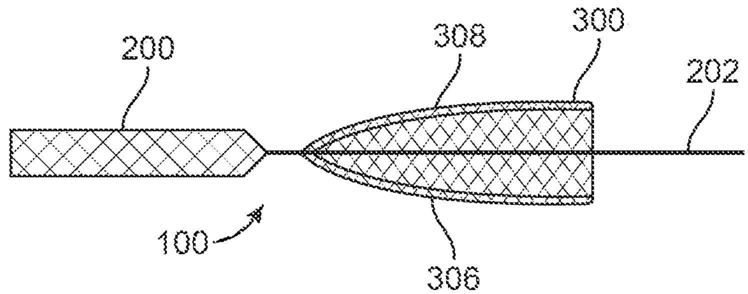


FIG. 5A

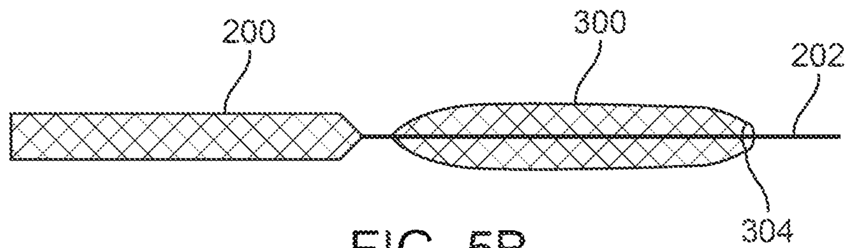


FIG. 5B

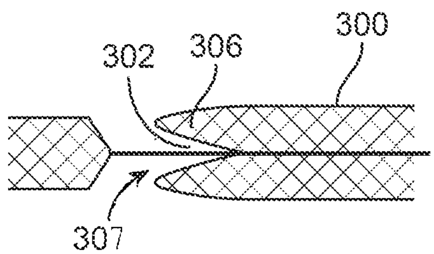


FIG. 5C

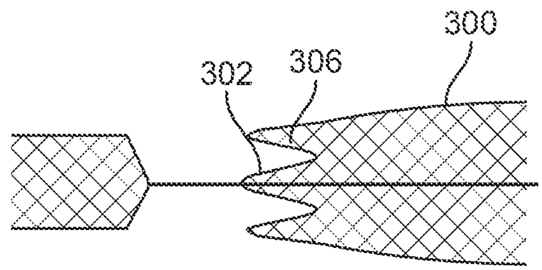


FIG. 5D

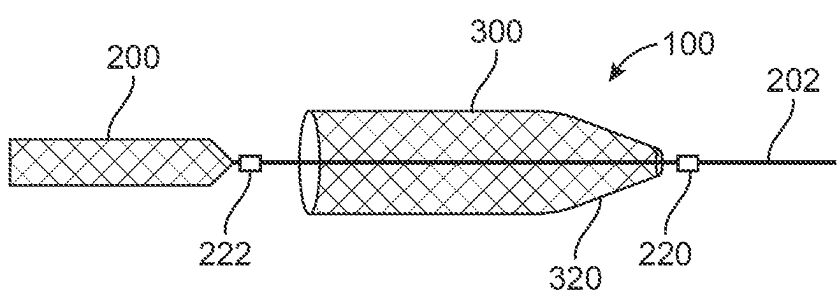


FIG. 5E

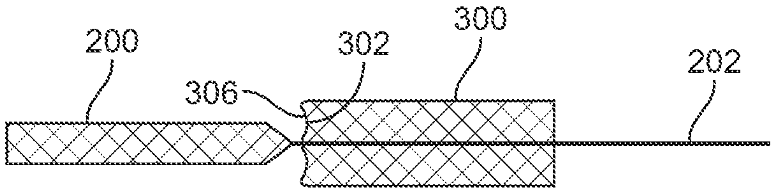


FIG. 5F

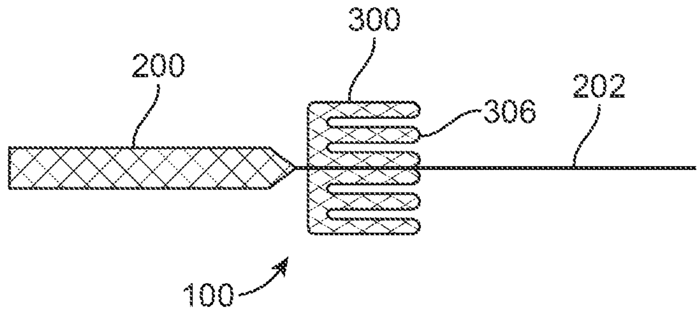


FIG. 5G

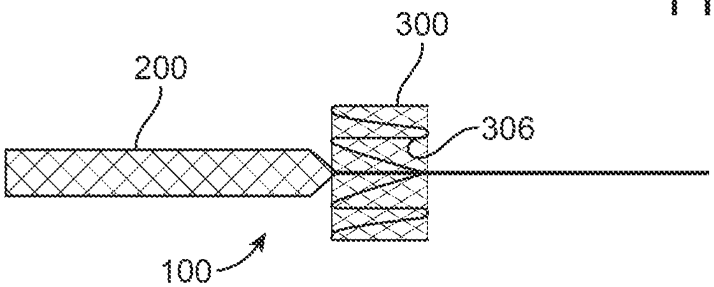
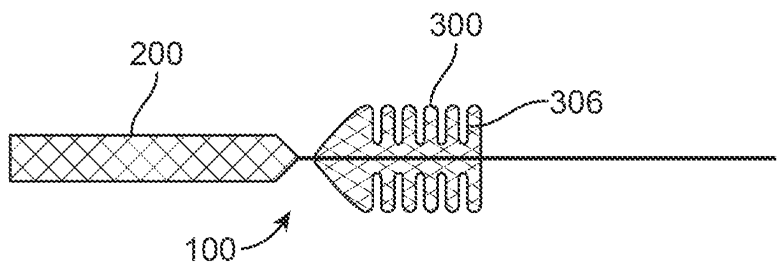


FIG. 5H



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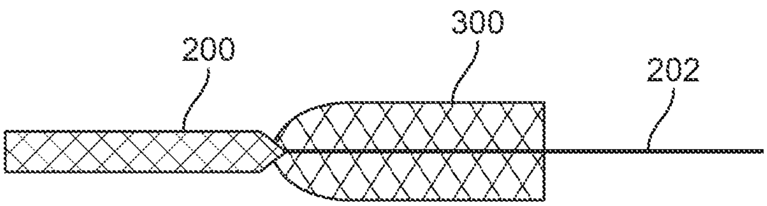

FIG. 5J

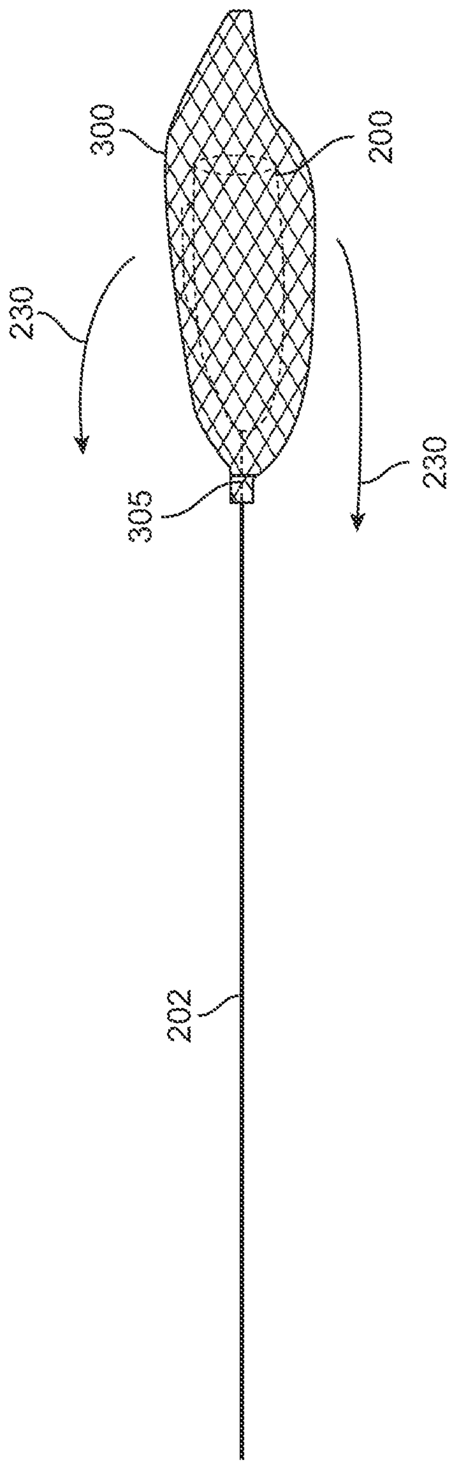

FIG. 5K

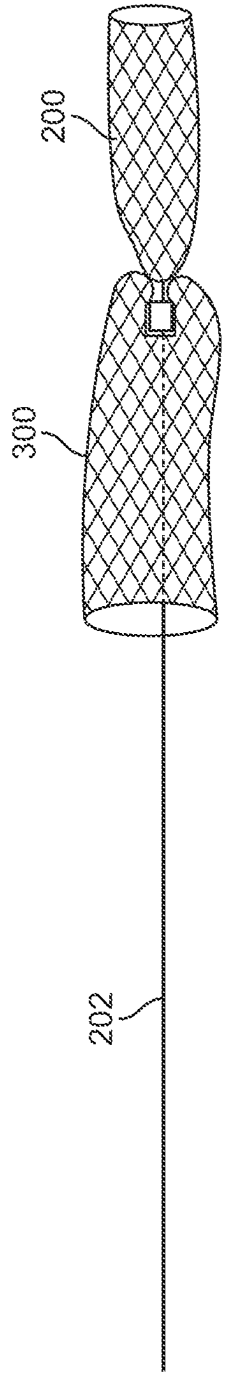

FIG. 5L

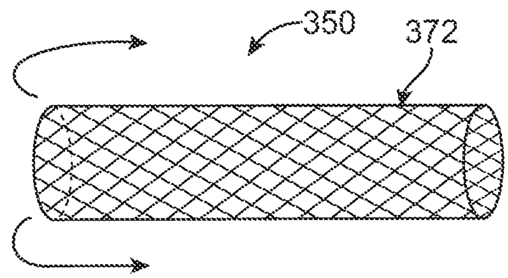

FIG. 6A

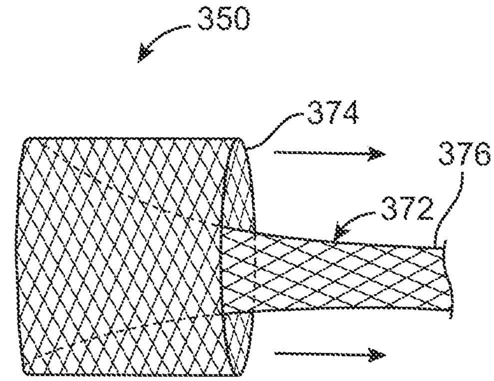

FIG. 6B

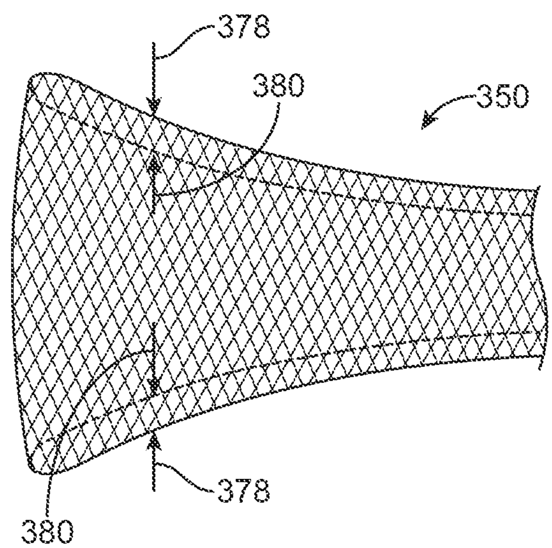

FIG. 6C

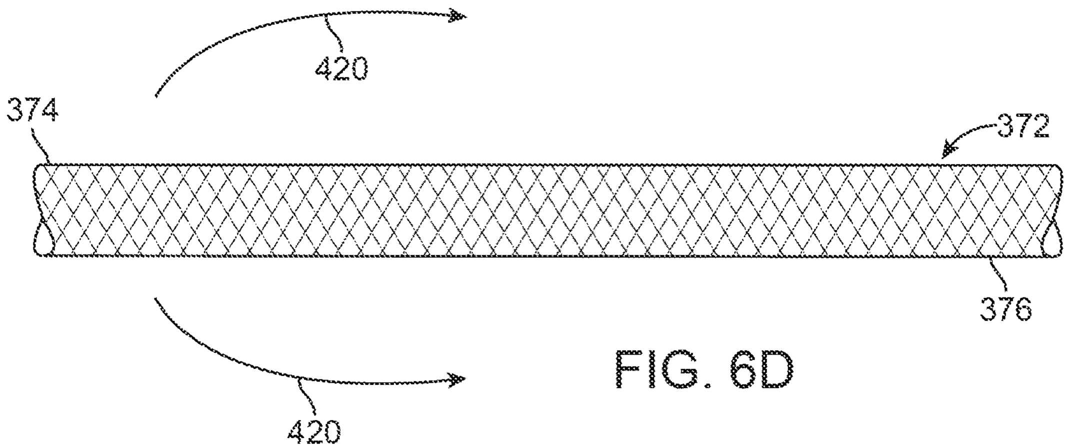

FIG. 6D

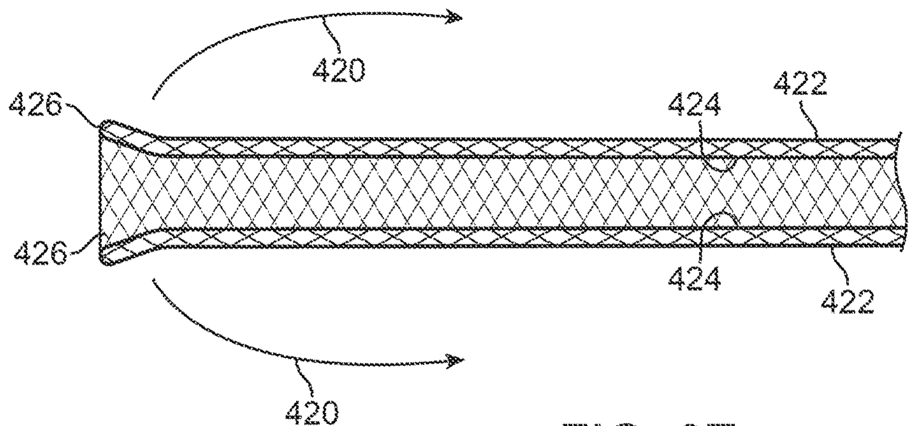

FIG. 6E

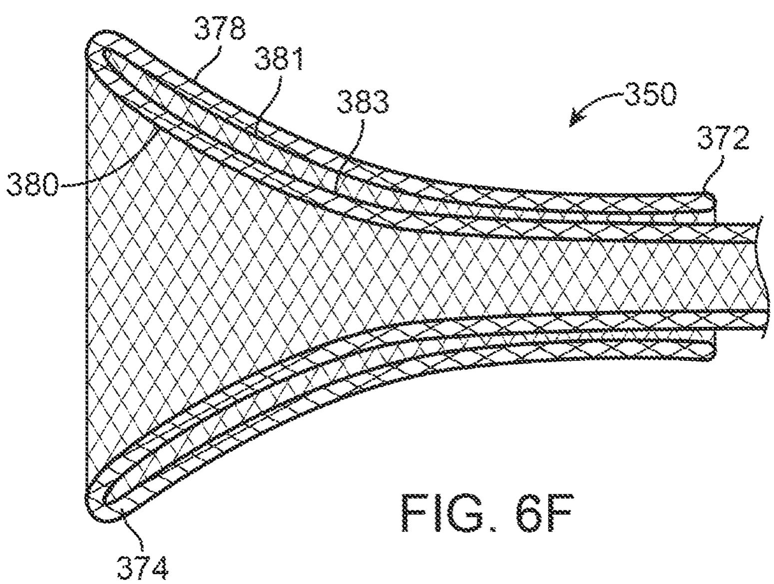

FIG. 6F

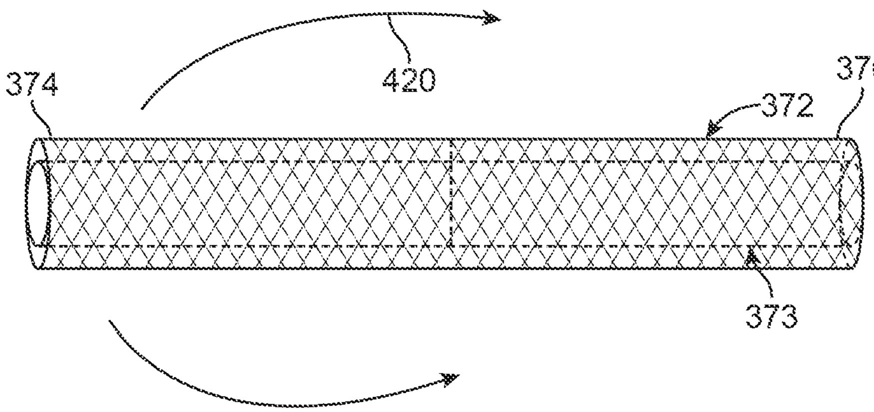

FIG. 6G

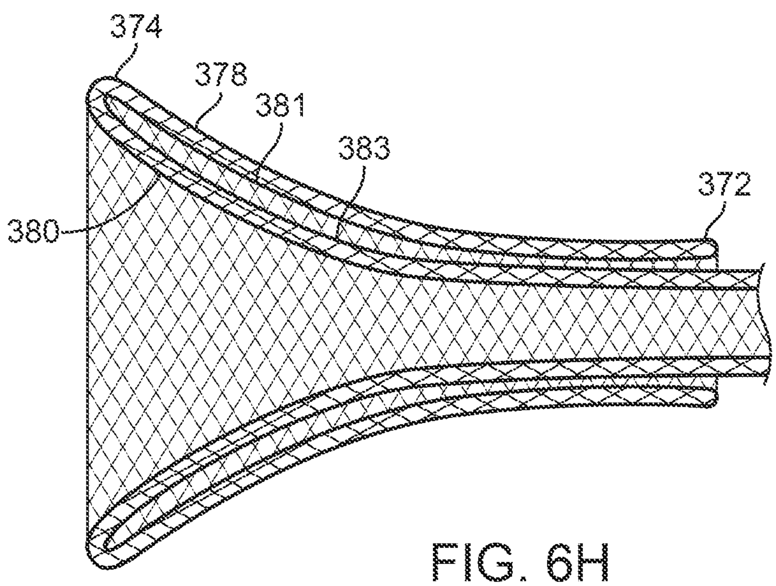

FIG. 6H

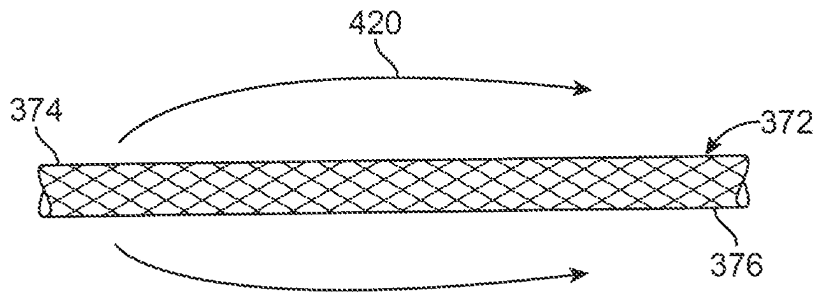

FIG. 6I

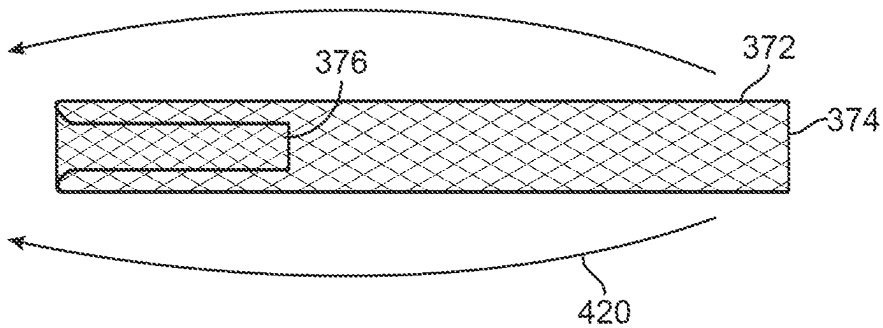

FIG. 6J

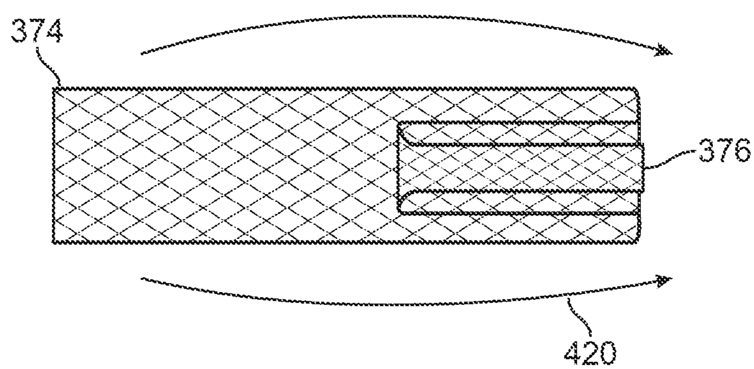

FIG. 6K

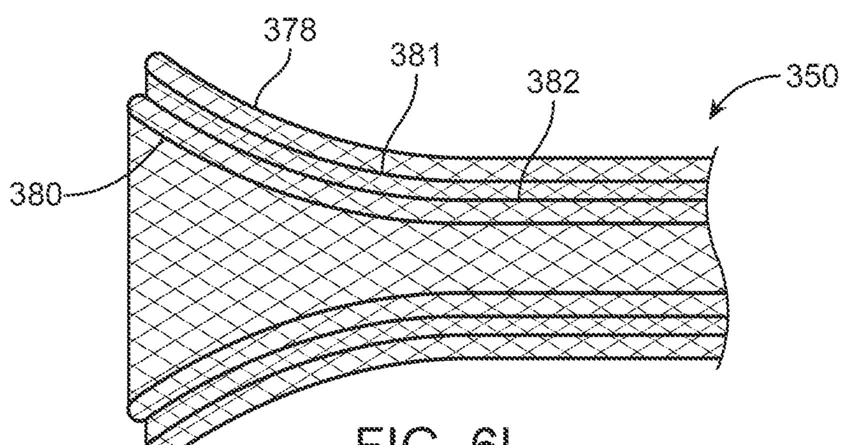

FIG. 6L

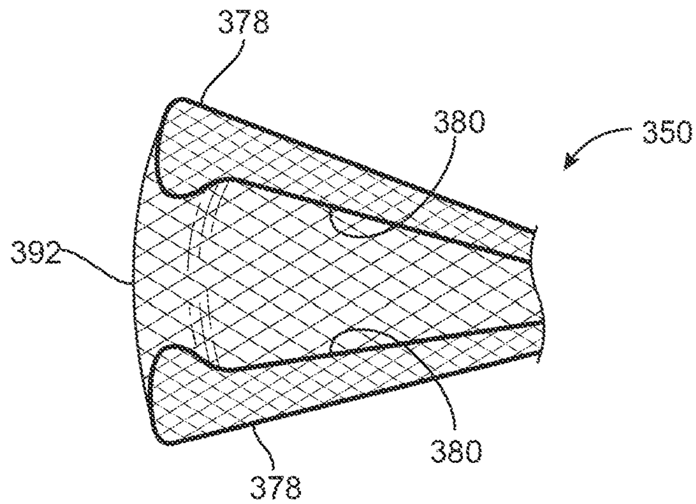

FIG. 7A

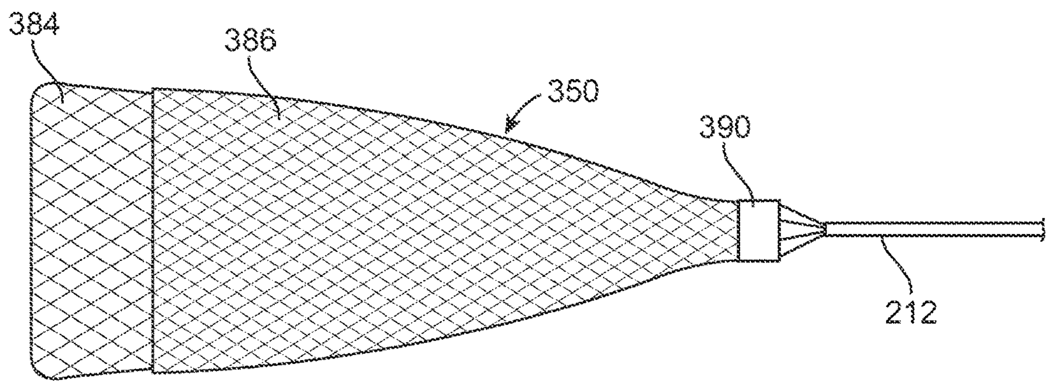

FIG. 7B

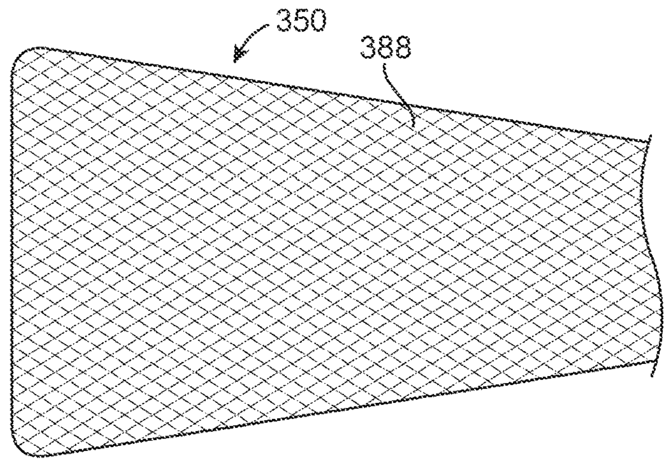

FIG. 7C

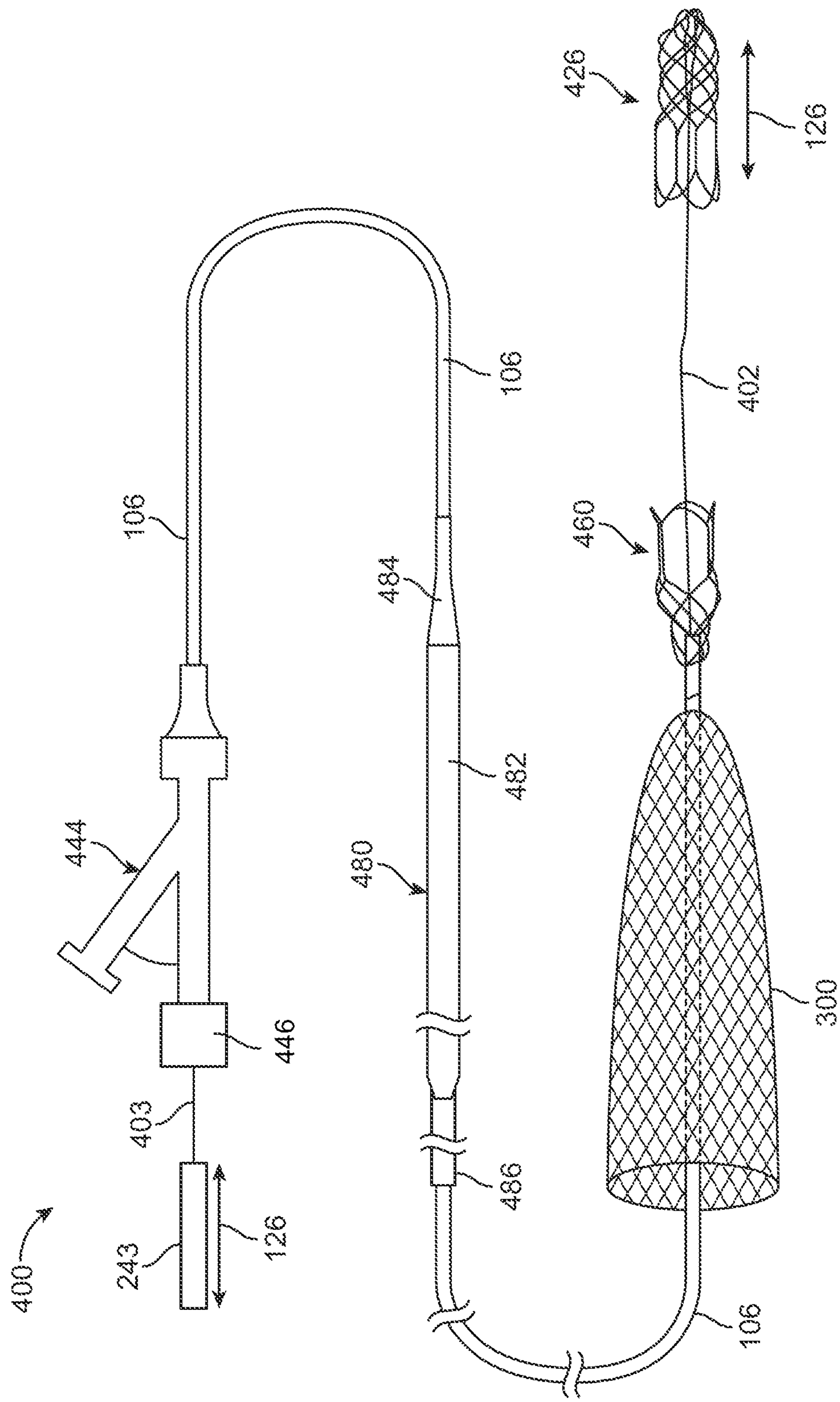
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FIG. 8

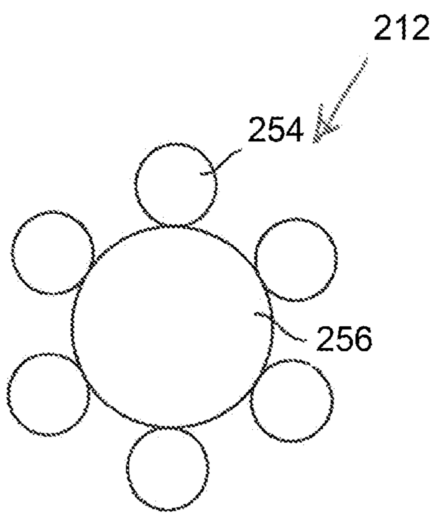

FIG. 9A

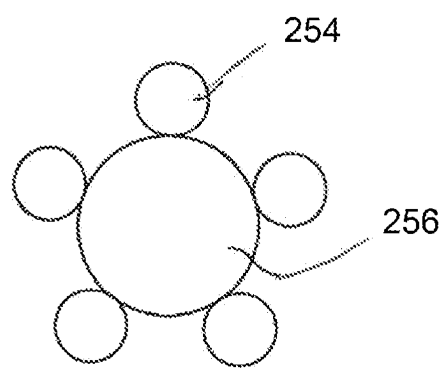

FIG. 9B

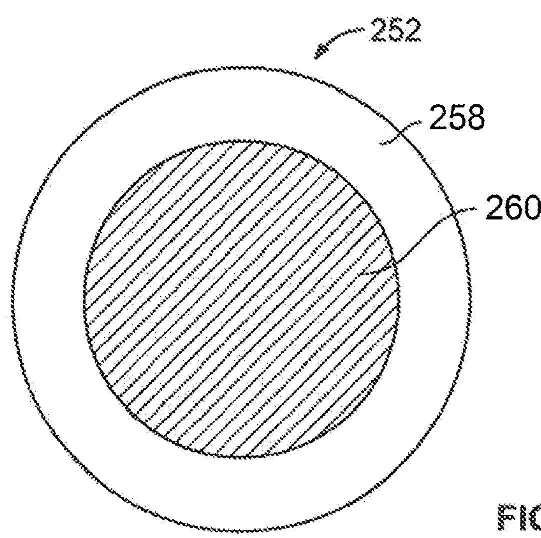

FIG. 9C

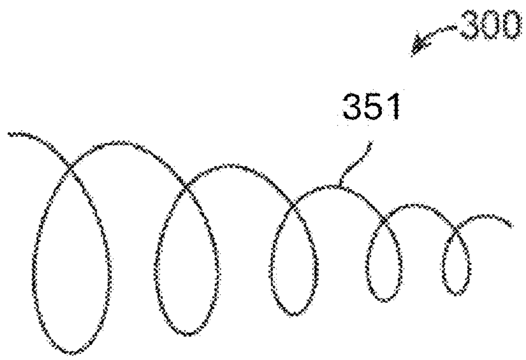

FIG. 10A

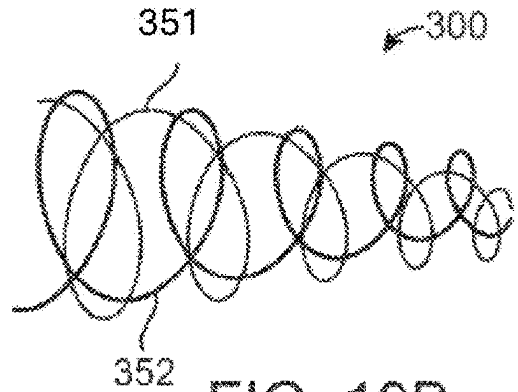

FIG. 10B

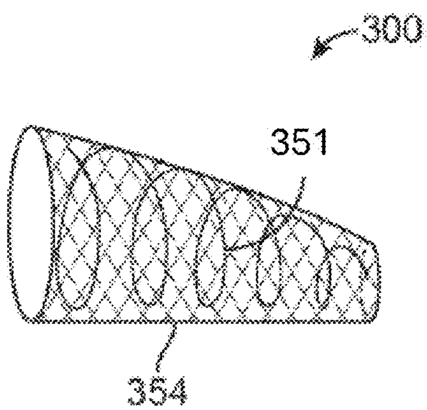

FIG. 10C

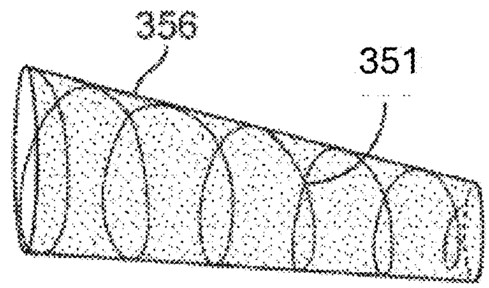

FIG. 10D

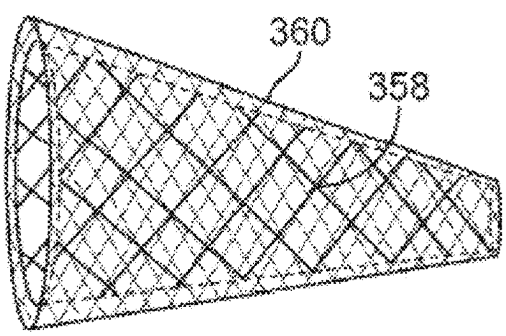

FIG. 10E

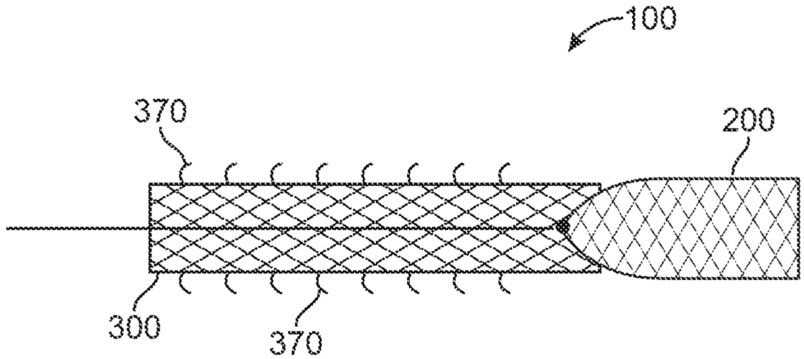

FIG. 11A

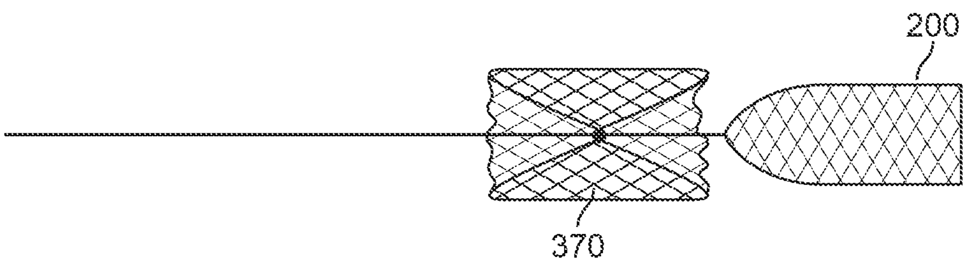

FIG. 11B

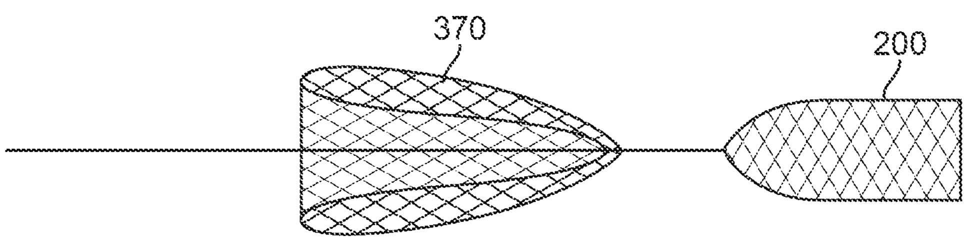

FIG. 11C

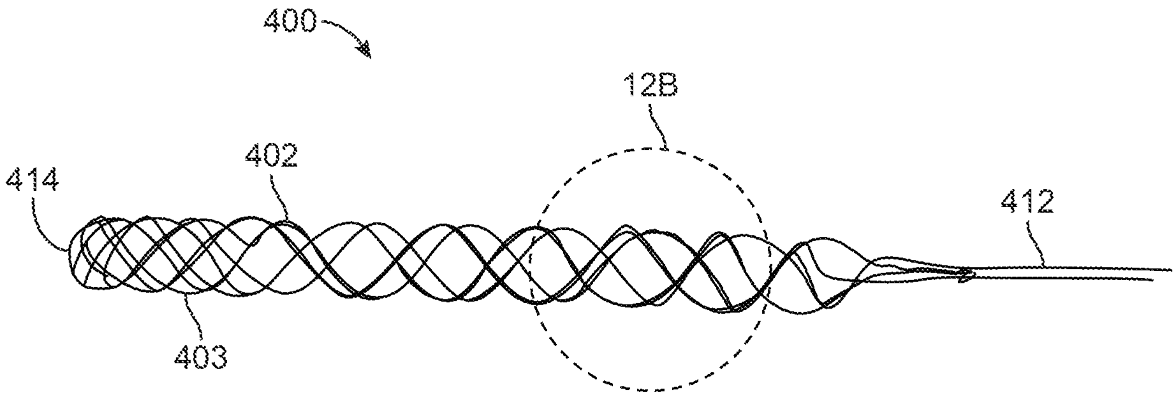

FIG. 12A

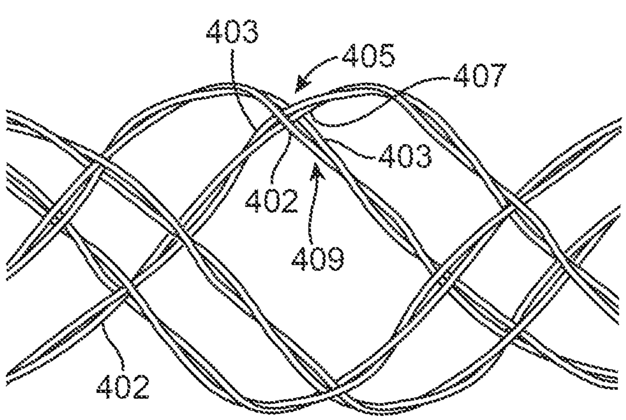

FIG. 12B

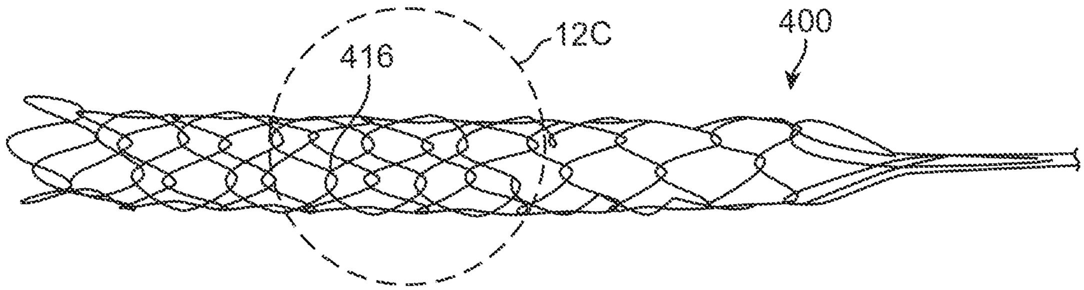

FIG. 12C

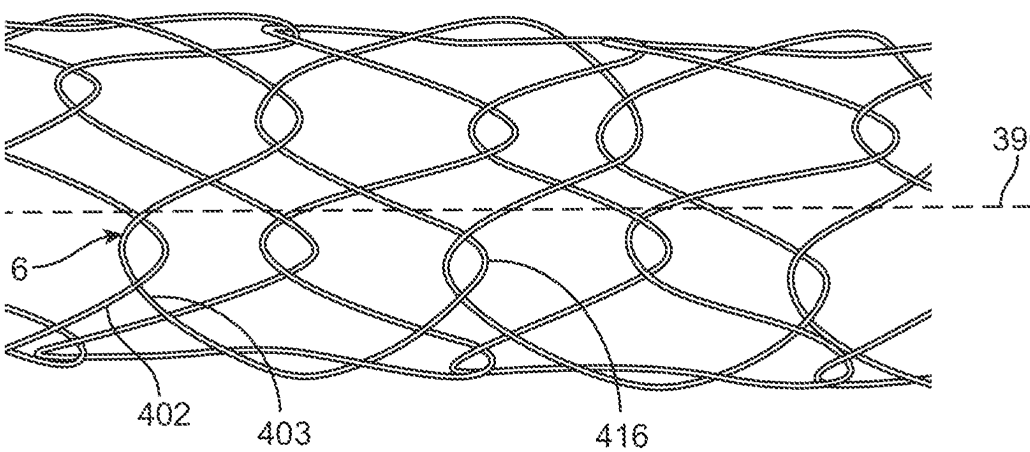

FIG. 12D

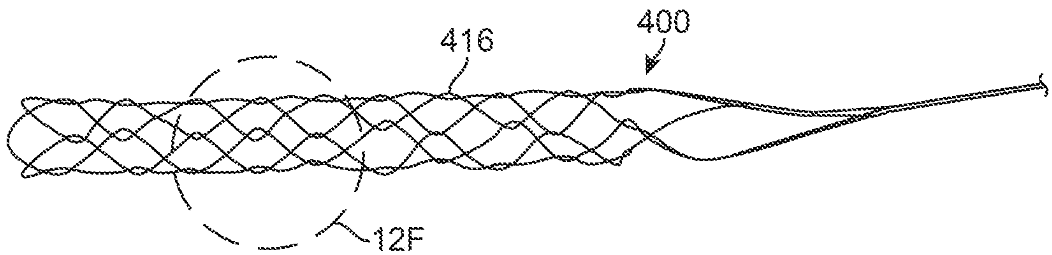

FIG. 12E

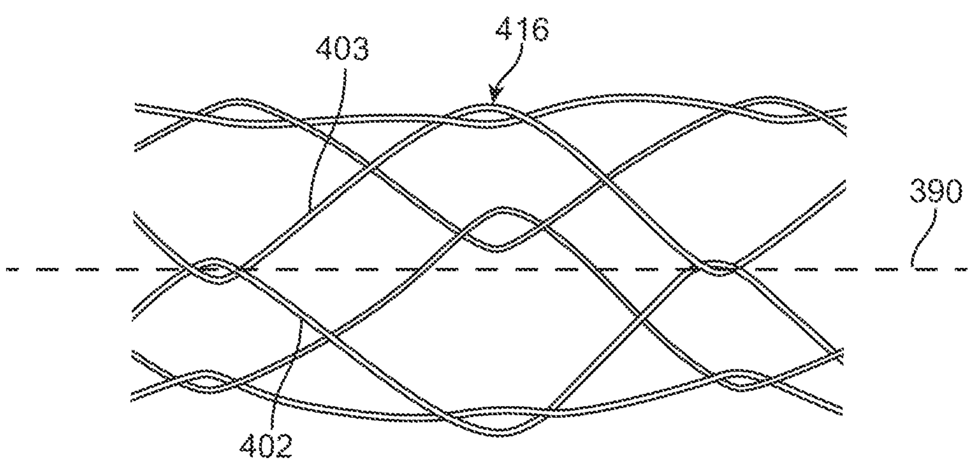

FIG. 12F

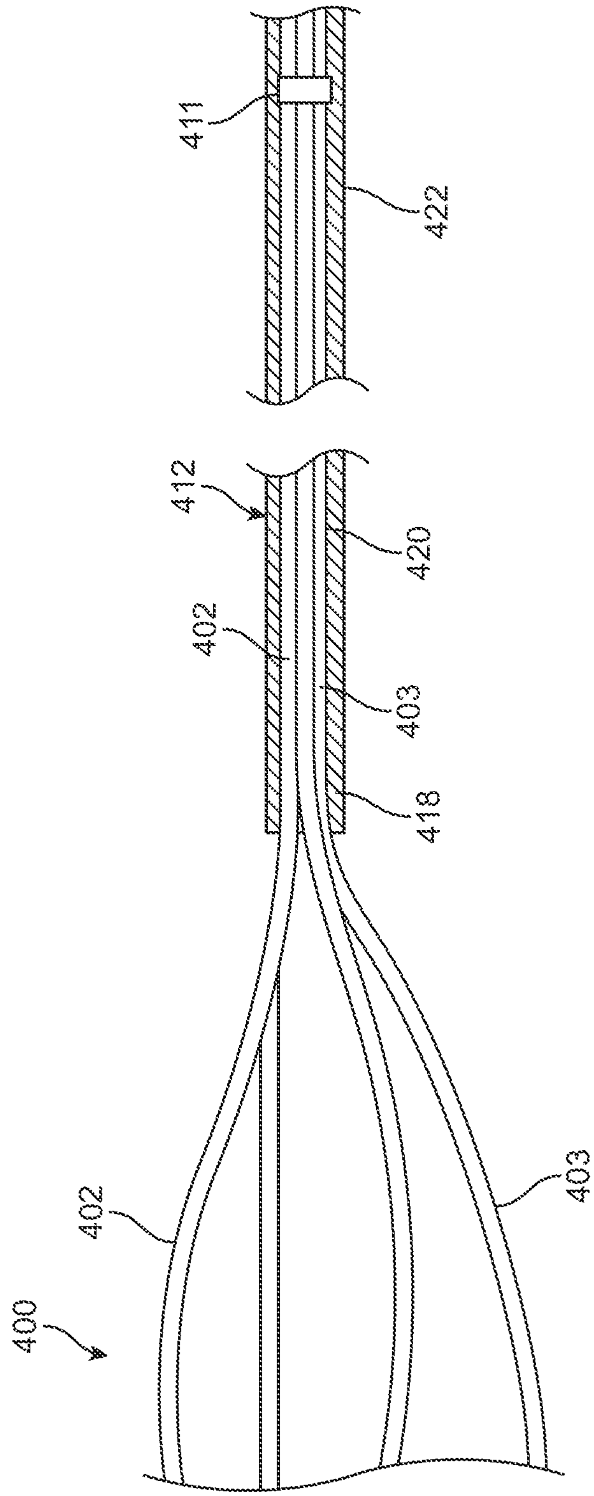

FIG. 12G

RETRIEVAL SYSTEMS AND METHODS FOR USE THEREOF

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is a continuation of U.S. patent application Ser. No. 16/530,101, filed Aug. 2, 2019, which is a continuation of U.S. patent application Ser. No. 15/946,466, filed Apr. 5, 2018, now U.S. Pat. No. 11,213,307, which is a continuation of U.S. patent application Ser. No. 15/174,016, filed Jun. 6, 2016, now U.S. Pat. No. 9,943,323, which is a continuation of U.S. patent application Ser. No. 14/446,755, filed Jul. 30, 2014, now U.S. Pat. No. 9,358,094, which is a continuation of U.S. patent application Ser. No. 13/959,433, filed Aug. 5, 2013, now U.S. Pat. No. 8,795,305, which is a continuation of PCT Application No. PCT/US2012/039216, filed May 23, 2012, which is a non-provisional of U.S. Provisional Application No. 61/489,183, filed May 23, 2011, and U.S. Provisional Application No. 61/489,254, filed May 24, 2011. The entirety of each of the above-mentioned prior applications is hereby incorporated by reference.

FIELD OF THE INVENTION

The devices described herein are intended to retrieve obstructions from the body. Such devices have applicability throughout the body, including clearing of blockages within body lumens and providing passive protection of such, such as the vasculature, by providing a capturing portion that can translate and/or mobilize the obstruction within the body lumen.

BACKGROUND OF THE INVENTION

A large number of medical procedures require the use of medical device(s) to remove an obstruction from a body lumen, vessel, or other organ. An inherent risk in such procedures is that mobilizing or otherwise disturbing the obstruction can potentially create further harm if the obstruction or a fragment thereof dislodges from the retrieval device. If a particle or the obstruction breaks free from the device and flows downstream, it is highly likely that the particle or obstruction will become trapped in smaller and more tortuous anatomy. In many cases, the physician will no longer be able to use the same retrieval device to again remove the obstruction because the size of the device may prevent advancing the device to the site of the new obstruction.

Even in successful procedures, a physician must proceed with caution to prevent the walls of the vessel or body lumen from imparting undesired forces to shear or dislodge the obstruction as it is translated through the body during removal. These forces have the potential of breaking portions or fragments of the obstruction away. In some cases, the obstruction can simply break free from the retrieval device and can lodge in a new area causing more concern than the original blockage.

Procedures for restoring flow within the cerebral vasculature as a result of ischemic stroke are one example of where these issues present a concern. The brain relies on its arteries and veins to supply oxygenated blood from the heart and lungs and to remove carbon dioxide and cellular waste from brain tissue. Blockages that interfere with this supply eventually cause the brain tissue to stop functioning. If the disruption in supply occurs for a sufficient amount of time,

the continued lack of nutrients and oxygen causes irreversible cell death (infarction). Accordingly, immediate medical treatment of an ischemic stroke is critical for the recovery of a patient. To access the cerebral vasculature a physician typically advances a catheter from a remote part of the body (typically a leg) through the vasculature and into the cerebral region of the vasculature. Once within the cerebral region, the physician deploys a device for retrieval of the obstruction causing the blockage. Concerns about dislodged obstructions or the migration of dislodged fragments increases the duration of the procedure at time when restoration of blood flow is paramount. Furthermore, a physician might be unaware of one or more fragments that dislodge from the initial obstruction and cause blockage of smaller more distal vessels.

Many physicians currently use stents to perform thrombectomy (i.e. clot removal) to resolve ischemic stroke. Typically, the physician deploys the stent into the clot, in an attempt to push the clot to the side of the vessel and re-establish blood to flow. Tissue plasminogen activator ("Tpa") is often injected into the bloodstream through an intravenous line. The TPA must travel in the blood stream until it reaches the clot that is causing the blockage. Once the Tpa contacts the clot, it begins to break up the clot with the hope of restoring blood flow to the affected areas. Tpa is also often administered to supplement the effectiveness of the stent. Yet, if attempts at clot dissolution are ineffective or incomplete, the physician can attempt to remove the stent while it is expanded against or enmeshed within the clot. In doing so, the physician must effectively drag the clot from the vessel, in a proximal direction, into a guide catheter located within vessels in the patients neck (typically the carotid artery). While this procedure has been shown to be effective in the clinic and easy for the physician to perform, there remain some distinct disadvantages using this approach.

The stent may not sufficiently hold onto the clot as it drags the clot to the catheter. In such a case, the clot might not move from the vessel. Another risk is that use of the stent might mobilize the clot might from the original blockage site, but the clot might not adhere to the stent during translation toward the catheter. This is a particular risk when translating through bifurcations and tortuous anatomy. Furthermore, blood flow can migrate the clot (or fragments of the clot) into a branching vessel at a bifurcation. If the clot is successfully brought to the guide catheter in the carotid artery, yet another risk is that the clot may be "stripped" or "sheared" from the stent as the stent enters the guide catheter. Regardless, simply dragging an expanded stent (either fully or partially expanded) can result in undesired trauma to the vessel. In most cases, the stent is oversized compared to the vessel. Dragging a fixed metallic (or other) structure can pull the arteries and/or strip the cellular lining from the vessel, causing further trauma such as a hemorrhagic stroke (leakage of blood from a cerebral vessel). Also, the stent can become lodged on plaque on the vessel walls resulting in further vascular damage.

In view of the above, there remains a need for improved devices and methods that can remove occlusions from body lumens and/or vessels. While the discussion focuses on applications in the cerebral vasculature, the improved devices and methods described below have applications outside of the area of ischemic stroke.

SUMMARY

The examples discussed herein show the inventive device in a form that is suitable to retrieve obstructions or clots within the vasculature. The term obstructions may include blood clot, plaque, cholesterol, thrombus, naturally occurring foreign bodies (i.e., a part of the body that is lodged within the lumen), a non-naturally occurring foreign body (i.e., a portion of a medical device or other non-naturally occurring substance lodged within the lumen.) However, the devices are not limited to such applications and can apply to any number of medical applications where elimination or reduction of the number of connection points is desired.

The devices discussed herein include interventional medical devices for retrieving and securing an obstruction within a vessel lumen. In one variation, the device includes a shaft having a flexibility to navigate through tortuous anatomy, the shaft having a distal portion and a proximal portion and a lumen extending therethrough; and an eversible cover having a fixed section affixed near an end of the distal portion of the shaft, a free section extending in a proximal direction from the fixed section and a cover wall extending from the fixed section to the free section, where the eversible cover is expandable against a vessel wall, the eversible cover being axially compliant such that when the interventional vascular device retrieval device is positioned through the shaft lumen and moved in a proximal direction, the friction of the eversible cover against the vessel wall causes the eversible cover to evert over the interventional vascular device allowing for the free section of the cover to be distal to the interventional vascular device.

In another variation, the present disclosure includes a method of securing an obstruction within a vessel. For example, the method can include advancing a shaft having a retrieval device affixed thereto to the obstruction; advancing a protective device over the shaft, the protective device comprising a sheath having an eversible cover, where a fixed end of the eversible cover is affixed to a distal portion of the sheath and a free end of the eversible cover is located proximal to the fixed end; positioning the fixed end of the eversible cover adjacent to the retrieval device and expanding at least a portion of the eversible cover against a portion of a wall of the vessel; proximally translating the shaft and retrieval device with at least a portion of the obstruction affixed thereto such that resistance of the eversible cover against the vessel resists movement of the eversible cover causing the free section of the eversible cover to evert over the proximally translated retrieval device.

Another variation of the method include securing an obstruction within a vessel of a patient, by providing an interventional vascular device having a wire attached thereto, where the interventional vascular device is configured to remove the obstruction from the vessel; coupling a shaft to the wire of the interventional vascular device, the shaft having a distal portion and a proximal portion and a lumen extending therethrough and having a flexibility to navigate through tortuous anatomy, an eversible cover having a fixed section affixed near an end of the distal portion of the shaft, a free section extending in a proximal direction from the fixed section and a cover wall extending from the fixed section to the free section, where the eversible cover is expandable against a vessel wall, the eversible cover being axially compliant such that when the interventional vascular device retrieval device is positioned through the shaft lumen and moved in a proximal direction against the eversible cover, the friction of the eversible cover against the vessel wall causes the eversible cover to evert over the interventional vascular device allowing for the free section of the cover to be distal to the interventional vascular device; positioning the shaft and eversible cover over the wire of the interventional vascular device prior to insertion into the patient; and advancing the interventional vascular device, shaft and eversible cover into the vessel to the obstruction. The device can also be configured so that the fixed section of the eversible cover comprises a pre-set shape to reduce a force required to evert the evertable cover.

The retrieval devices can comprise any number of capturing or retrieval device such as a filter, an atherectomy device, a rotational cutter, an aspiration device, stent based retrievers and retrieval baskets.

The methods described herein can include methods of securing an obstruction within a vessel. In one example, the method can comprise: positioning a catheter within a vessel; advancing a shaft having a retrieval device affixed thereto out of the catheter; advancing an eversible cover out of the catheter such that a fixed end of the eversible cover is affixed adjacent to a proximal end of the retrieval device and a free end of the eversible cover is moveable relative to the shaft and retrieval device; expanding a at least a portion of the eversible cover against a portion of a wall of the vessel; manipulating the retrieval device to become at least partially enmeshed with the obstruction; and proximally translating the shaft and retrieval device with at least a portion of the obstruction affixed thereto such that resistance of the eversible cover against the vessel resists movement of the eversible cover causing the free section of the eversible cover to evert over the proximally translated retrieval device.

In another variation, the methods can include further withdrawing the shaft from the vessel such that during withdrawal the eversible cover forms a protective barrier over the obstruction to lessen shearing forces caused by the vessel and reduce dislodging portions of the obstruction from the retrieval device.

Another variation of a method includes a method of preparing a retrieval device comprising: providing a retrieval device having been previously removed from a body of a patient where the retrieval device includes a protective cover where a fixed end of the protective cover is affixed adjacent to a proximal end of the retrieval device and where a free end is located distally to the fixed end covering the retrieval device and is moveable relative to the second end; reversing the protective cover by moving the free end proximally of the fixed while the fixed end remains affixed adjacent to the proximal end of the retrieval device; inserting the retrieval device and cover into a catheter where the free end of the cover is proximal to the fixed end of the cover and retrieval device such that upon deployment from the catheter, the free end of the cover deploys proximally to the fixed end of the cover.

In another example, the devices described herein can include medical device retrieval systems for securing an obstruction within a vessel lumen and for use with a catheter configured to be navigated through the vasculature. In one variation, the device comprises an elongated stent comprising a plurality of struts, the stent being collapsible for positioning in the catheter during delivery and having an expanded profile such that when expanded the struts are configured to engage the obstruction; a shaft fixedly attached to the elongated stent and having a flexibility to navigate through tortuous anatomy; a fluid permeable cover having a distal end coupled to a proximal end of the elongated stent a cover wall defining a cavity and extending along the shaft, and a proximal end being moveable relative to the shaft, where the fluid permeable cover is collapsible for positioning in the catheter during delivery and is expandable upon deployment from the catheter such that at least a portion of the fluid permeable cover is expandable; where the fluid permeable cover is axially pliable such that when the device is deployed in the vessel the frictional forces between the vessel and the fluid permeable cover permit proximal movement of the shaft and elongated stent to cause inversion of the fluid permeable cover wall such that the fluid permeable cover wall everts over the elongated stent.

Another variation of a device includes an interventional medical device for use with a catheter configured for delivery through vasculature for securing an obstruction within a vessel lumen. For example, the device can comprise a shaft having a flexibility to navigate through tortuous anatomy, the shaft having a distal portion and a proximal portion; a capturing device comprising a sidewall, the capturing device fixedly located at a distal portion of the shaft and having a reduced profile for positioning in the catheter and an expanded profile, such that upon deployment from the catheter, the capturing device expands to force a portion of the sidewall into the obstruction to at least partially attach to the obstruction; a cover having a distal end coupled adjacent to a proximal end of the capturing structure, a proximal end and a cover wall extending therebetween, where the proximal end of the cover is slidable relative to the distal end, where the cover is expandable such that when located in the catheter the cover is in a reduced delivery state and upon advancement from the catheter the cover expands with the proximal end located proximally of the distal end, where the cover wall is compliant such that when deployed from the catheter and the shaft is pulled in a proximal direction frictional forces between the vessel and the cover wall or proximal end cause the cover to invert as the cover wall inverts over the capturing device to surround the capturing device.

Another variation of the device includes an interventional medical device for securing a retrieval device having one or more obstructions located therein for removal from a body. In one such example the medical device includes a sheath having a flexibility to navigate through tortuous anatomy, the sheath a distal portion and a proximal portion and a lumen extending therethrough; an eversible cover having a fixed section affixed to the distal portion of the sheath, a free section extending in a proximal direction from the fixed section and a cover wall extending from the fixed section to the free section, where the eversible cover is expandable, the eversible cover being axially compliant such that when the retrieval device is positioned through the sheath lumen moved in a proximal direction against the eversible cover, the eversible cover everts over the retrieval device allowing for the free section of the cover to be distal to the retrieval device.

Another variation of the method includes advancing a shaft having a retrieval device affixed thereto to the obstruction; advancing a protective device over the shaft, the protective device comprising a sheath having an eversible cover, where a fixed end of the eversible cover is affixed to a distal portion of the sheath and a free end of the eversible cover is located proximal to the fixed end; positioning the fixed end of the eversible cover adjacent to the retrieval device and expanding at least a portion of the eversible cover against a portion of a wall of the vessel; proximally translating the shaft and retrieval device with at least a portion of the obstruction affixed thereto such that resistance of the eversible cover against the vessel resists movement of the eversible cover causing the free section of the eversible cover to evert over the proximally translated retrieval device.

The capturing portions described herein can include a stent retrieval device for expanding against one or more occlusive bodies in a vasculature. In one example, the stent retrieval device includes an elongate shaft having a flexibility to navigate through tortuous anatomy, the elongate shaft having a distal portion and a proximal portion; a plurality of filaments that diverge from the distal portion of the elongate shaft to form an expandable elongated stent body having a open distal end and a fluid permeable closed proximal end and a cavity therebetween, where divergence of the filaments at the distal portion of the elongate shaft forms the fluid permeable closed proximal end; where the plurality of filaments extending along the shaft are free of any connection joints in the distal portion to permit increased flexibility of the distal portion as it navigates though tortuous anatomy; and one or more connection joints proximal to the distal portion where the connection joints secure the plurality of filaments to the shaft.

The stent retrieval can also include at least one of the plurality of filaments that comprise at least two wires twisted together, the elongated stent body further comprising at least one intersection of filaments, where the wires of each filament are interwoven to provide increased outward radial strength of the elongated stent body and such that the wires slide relative to each other as the elongated stent body expands or compresses in diameter to reduce a force required to linearize the elongated stent body.

The stent retrieval device can have an exterior surface of the elongated stent body that comprises an irregular surface formed by intersection of filaments.

The stent retrieval device can also have intersection of filaments comprising a barb or knuckle and where a plurality of barbs or knuckles is radially spaced about the elongated stent body. The stent retrieval device can also have an intersection of filaments that comprises a barb or knuckle and where a plurality of barbs or knuckles is aligned with an axis of the elongated stent body.

In one variation of the devices described herein, the device comprises a main bundle or group of wires that diverge to form a device having various shapes but few or no connections points or joints (where fabrication of such a construction is referred to as "jointless"). Clearly, the inventive devices described herein are not limited to such a jointless construction. Additional variation includes one or more leading wires that are attached to a capturing portion as described below.

Devices of the present invention can incorporate any number of wires of different characteristics including, but not limited to, materials, shapes, sizes and/or diameters. Clearly, the number of permutations of device configurations is significant. Providing devices with such a composite construction allows for the manipulation of the device's properties to suite the intended application.

As noted herein, the joint-less construction improves the flexibility and strength of the device by eliminating joints, connection points, or other attachment points. In addition, the joint-less construction improves the ability of the device to be delivered through a small microcatheter. As a result, the device and microcatheter are able to access remote regions of the vasculature.

The devices may be fabricated to be self-expanding upon deployment from a catheter. Alternatively, the devices can be constructed from shape-memory alloys such that they automatically deploy upon reaching a pre-determined transition temperature.

It should be noted that in some variations of the invention, all or some of the device can be designed to increase their ability to adhere to the obstruction. For example, the wires may be coupled to an energy source (e.g., RF, ultrasonic, or thermal energy) to "weld" to the obstruction. Application of energy to the device can allow the surrounding portion to deform into the obstruction and "embed" within the obstruction. Alternatively, the device can impart a positive charge to the obstruction to partially liquefy the obstruction sufficiently to allow for easier removal. In another variation, a negative charge could be applied to further build thrombus and nest the device for better pulling force. The wires can be made stickier by use of a hydrophilic substance(s), or by chemicals that would generate a chemical bond to the surface of the obstruction. Alternatively, the filaments may reduce the temperature of the obstruction to congeal or adhere to the obstruction.

Additional devices and methods for treating ischemic stroke are discussed in commonly assigned U.S. patent application Ser. No. 11/671,450 filed Feb. 5, 2007; Ser. No. 11/684,521 filed Mar. 9, 2007; Ser. No. 11/684,535 filed Mar. 9, 2007; Ser. No. 11/684,541 filed Mar. 9, 2007; Ser. No. 11/684,546 filed Mar. 9, 2007; Ser. No. 11/684,982 filed Mar. 12, 2007, Ser. No. 11/736,526 filed Apr. 17, 2007, Ser. No. 11/736,537 filed Apr. 17, 2007, Ser. No. 11/825,975 filed Sep. 10, 2007; Ser. No. 12/344,378 filed Dec. 26, 2008, Ser. No. 13/012,727 filed Jan. 24, 2011, and Ser. No. 13/226,222 filed Sep. 6, 2011; the entirety of each of which is incorporated by reference. The principles of the invention as discussed herein may be applied to the above referenced cases to produce devices useful in treating ischemic stroke. In other words, the wire-shaped construction of devices according to present invention may assume the shapes disclosed in the above-referenced cases when such a combination is not inconsistent with the features described herein.

BRIEF DESCRIPTION OF THE DRAWINGS

Each of the following figures diagrammatically illustrates aspects of the invention. Variation of the invention from the aspects shown in the figures is contemplated.

FIG. **1** illustrates an example of a device according to the present invention when used in a system for removing obstructions from body lumens.

FIGS. **2**A to **2**C illustrate working ends of various coverable retrieval devices.

FIGS. **2**D and **2**E show variations of retrieval devices.

FIG. **2**F shows an independent eversible cover on a delivery sheath.

FIGS. **3**A to **3**C illustrate an example of a coverable retrieval device where the cover everts about the retrieval structure.

FIG. **4**A to **4**I illustrates an example where an improved retrieval device with passive protection retrieves a clot from tortuous anatomy.

FIGS. **4**J and **4**K illustrate examples of an obstruction or other material captured within a retrieval device with a cover further protecting the loaded retrieval device.

FIG. **5**A illustrates a retrieval device having a retrieval structure adjacent to a double layer cover.

FIG. **5**B shows a funnel with a free end that tapers down about the delivery wire.

FIGS. **5**C and **5**D show a fixed end of a cover that is pre-shaped to reduce the force required to evert the cover wall.

FIG. **5**E shows alternate variation of a passive cover integrated into a retrieval device.

FIG. **5**F illustrates a cover having a pre-set flattened cover wall at a fixed end of the retrieval structure.

FIGS. **5**G to **5**I illustrate various layered covers.

FIG. **5**J shows a cover that is constructed directly onto the retrieval structure rather than the delivery shaft.

FIGS. **5**K and **5**L show a variation of a cover and retrieval device where the cover is first mounted in a distal direction and then inverted in a proximal direction.

FIGS. **6**A to **6**L illustrate a variation of covers for use as describe herein.

FIGS. **7**A to **7**C show additional variations of covers.

FIG. **8** illustrates a variation of a proximal and distal end of an additional retrieval device.

FIGS. **9**A to **9**C illustrate wires of different constructions within a delivery wire or shaft.

FIGS. **10**A to **10**E illustrate additional variations of covers for use as described above.

FIGS. **11**A to **11**C illustrate additional variations of covers for use with the devices and methods described herein.

FIGS. **12**A to **12**F illustrate various stent designs for increasing the ability of a stent to adhere to an occlusion within a vessel.

FIG. **12**G illustrates a proximal end of the stent structure.

DETAILED DESCRIPTION

It is understood that the examples below discuss uses in the cerebral vasculature (namely the arteries). However, unless specifically noted, variations of the device and method are not limited to use in the cerebral vasculature. Instead, the invention may have applicability in various parts of the body. Moreover, the invention may be used in various procedures where the benefits of the method and/or device are desired.

FIG. **1** illustrates a system **10** for removing obstructions from body lumens as described herein.

In the illustrated example, this variation of the system **10** is suited for removal of an obstruction in the cerebral vasculature. As stated herein, the present devices and methods are useful in other regions of the body including the vasculature and other body lumens or organs. For exemplary purposes, the discussion shall focus on uses of these devices and method in the vasculature.

It is noted that any number of catheters or microcatheters may be used to locate the catheter/microcatheter **12** carrying the obstruction removal device **200** at the desired target site. Such techniques are well understood standard interventional catheterization techniques. Furthermore, the catheter **12** may be coupled to auxiliary or support components **14**, **16** (e.g., energy controllers, power supplies, actuators for movement of the device(s), vacuum sources, inflation sources, sources for therapeutic substances, pressure monitoring, flow monitoring, various bio-chemical sensors, bio-chemical substance, etc.) Again, such components are within the scope of the system **10** described herein.

In addition, devices of the present invention may be packaged in kits including the components discussed above along with guiding catheters, various devices that assist in the stabilization or removal of the obstruction (e.g., proximal-assist devices that holds the proximal end of the obstruction in place preventing it from straying during removal or assisting in the removal of the obstruction), balloon-tipped guide catheters, dilators, etc.

FIG. **2**A illustrates a working end of a coverable retrieval device **100**. Typically, the device includes a capturing or retrieval structure **200**. In the illustrated example, the retrieval structure **200** comprises an elongated stent structure. However, unless specifically noted, the capturing structure can comprise any number of devices, including but not limited to a filter, an atherectomy device, a rotational cutter, an aspiration catheter.

The retrieval structure **200** is located at a distal end of a delivery wire **202**. In one variation, the retrieval structure **200** can be permanently affixed to the delivery wire **200** by such methods including, but not limited to adhesive bonding, soldering, welding, polymer joining, or any other conventional method. In some variations, the retrieval device **200** can be formed from one or more wires forming the delivery wire **202** or shaft **202**. The delivery wire **202** can have sufficient column strength such that it can axially advance and retract the device **100** within the vasculature as the physician manipulates a non-working end of the delivery wire **202** outside of the body. Accordingly, the delivery wire **202** should have a length that is sufficient to extend from the target area, e.g., the cerebral vasculature, to the entry point on the body. Alternatively, additional variations of the device **100** can allow for the use of a support member or catheter that positions the retrieval structure **200** as needed. Additional features of the retrieval structure **200** can be found in the commonly assigned patents and applications cited herein an incorporated by reference.

The coverable retrieval device **100** further includes a cover **300** (also referred to as a funnel or sheath) affixed relative to a proximal end **206** of the retrieval structure **200**. By being affixed relative to a proximal end **206**, a distal end **204** of the retrieval structure **200** can move relative to the cover **300** so that the cover **300** everts over the proximal end **206** of the structure **200** when the cover **300** is expanded within a vessel and as the structure **200** is withdrawn into the distal end **302** of the cover **300**. This mechanism is discussed in detail below.

FIGS. **2**B and **2**C illustrate alternative variations of a coverable retrieval device **100**. As shown in FIGS. **2**B and **2**C, the distal end **302** of the cover **300** can be spaced from the proximal end **206** of the retrieval structure **200**. Alternatively, the distal end **302** of the cover **300** can extend over a portion of the retrieval structure **200**. In some variations, at least a section of the cover **300** expands to a greater diameter than a diameter of the retrieval structure **200**. This allows the cover **300** to expand to a vessel wall where the vessel holds the cover stationary while the device is pulled proximally through the cover to evert the cover. In alternate variations, the cover **300** expands to the same or lesser diameter than the retrieval structure **200** or other device.

FIG. **2**D shows a retrieval device **100** with a catheter **112** (usually a microcatheter). The retrieval device **100** can comprise a single unitary device of a cover **300** and retrieval structure **200** (in this case the retrieval structure is an elongated stent structure). One benefit of a unitary device is that additional devices complicates the procedure and can increase the duration of what is ordinarily a time sensitive procedure. The retrieval device **100** can be positioned through the catheter **112** that includes a hub **114**. As a result, the physician only needs to manipulate the unitary retrieval device **100** and the catheter/microcatheter **112**. The retrieval device **100** is loaded into the catheter **112** for placement at the target site. In addition, the retrieval device can be reloaded if the procedure must be repeated. The cover **300** and retrieval structure **200** described herein can comprise any construction described herein or as known by those skilled in the art.

FIG. **2**E shows a retrieval device **100** with a cover **300** and retrieval device **200** with a radiopaque marker **305** therebetween. As shown, variations of the device **100** do not require a catheter or microcatheter.

FIG. **2**F illustrates an eversible cover **300** located on a sheath **330** having a lumen **332** extending therethrough. A separable retrieval device **200** can be coupled to the cover **300** and sheath **330** by inserting the wire **202** of the cover retrieval device **200** through the lumen **332** of the sheath **330**. In this variation, the eversible cover **300** can be used with any number of different interventional tools. The separate devices can be assembled prior to delivery into the patient. Alternatively, the devices can be positioned within the body and subsequently joined once the retrieval device **200** engages the target area.

FIG. **3**A illustrates an example of a coverable retrieval device **100** where the cover **300** is in the process of everting about the retrieval structure **200**. As shown, arrow **50** illustrates a force applied on the wire **202** in a proximal direction. Arrows **52** illustrate a resistance force applied by the friction of the expanded cover **300** against a vessel or similar wall. This friction force **52** prevents or resists proximal movement of the free end **304** of the cover **300** while the fixed end **302** moves in a proximal direction with the proximal end **206** of the retrieval structure **200**. This action causes a wall **306** of the cover **300** to evert over the retrieval structure **200**. Ultimately, and as shown in FIG. **3**B, the free end **304** of the cover **300** ends up distally over the fixed end **302**. As shown, the wall of the everted cover **300** provides a safety type cover for the retrieval device **200**. In additional variations, the fixed end **302** of the cover can actually be slidable or moveable along the delivery wire **202**. However, the similar principle as discussed above shall apply to cause everting of the cover **300** over the retrieval structure **200**.

FIG. **3**C illustrates another variation of a coverable retrieval device **100** after the cover **300** is everted about the retrieval structure **200**. In this variation, the free end **304** of the cover **300** ends up distally of the fixed end **302** and tapers or collapses towards the free end **304**. The cover **300** can be shape set so that prior to eversion the cover is as shown above where the forces acting on the cover wall **306** expand outwards, but after eversion the forces on the cover wall **306** cause the tapering or collapsing as shown in FIG. **3**C.

In accordance with the illustrations discussed above, the cover **300** can be made so that the cover wall **306** is atraumatic when dragged across a lumen wall. The cover can be manufactured from any number of materials including a fabric, a reinforced fabric, a braid, weave, or any such material that allows for expansion against a wall of the body lumen or vessel as well as to allow everting of a wall **306** of the cover over the retrieval device **200**. The cover wall **306** can also comprise combinations of these materials such as braids of polymer material with metal fibers, soft braids with coil reinforcements or various other combinations.

The cover wall can comprise a mesh that can include any medically acceptable materials such as a Nitinol braid. Furthermore, the mesh allows for flow through the vessel or lumen while expanded. However, additional variations of the device can include a solid layer of material substituted for the mesh. Moreover the cover can comprise any number of configurations. For example, the cover can comprise a single layer wall or a multi layer wall, the open end of the cover could be made to have terminated ends such as by using continuous wire loops formed during the braiding process. Alternatively, the ends can be cut and then terminated by encasing in polymer, laser welds, or by folding inward for a discreet length and then terminating

In one example, the cover **300** comprises a continuous wire construction as described in earlier commonly assigned patent applications incorporated by reference. In one variation the cover **300** comprises a finely braided wire, such as 48-96 wires of 0.0005" to 0.002" diameter fine Nitinol wire or similar. Additionally, the wire can comprise cobalt chromium, stainless steel, or similar, or drawn filled tube (dft) with platinum core. In additional variations, a flat wire or oval wire can be used. The wire does not need to be uniform. Instead, a number of different types of wires can be used. Some of the individual wires could be platinum alloys for added radiopacity.

FIG. **4**A to **4**I illustrates an example where an improved retrieval device **100** with passive protection retrieves a clot **2** from tortuous anatomy. FIG. **4**A illustrates a clot **2** that obstructs blood flow in a vessel **6**. As noted herein, the vessel **6** can comprise any vessel in cerebral vasculature, coronary or peripheral vasculature. Alternatively, the device and methods for use are not limited to use in the vasculature. Variations of the principles, concepts, method and devices described herein can be applicable wherever a retrieval device can be used. FIG. **4**A also illustrates a guide sheath or access catheter **108** that is advanced within the vessel. During a procedure, the physician will advance the access catheter **108** as far distally as possible. However, due to the size of the access catheter **108**, a physician typically positions it a distance away from the obstruction **2**. As shown, there can be any number of bifurcations **8** in the vessel **6** located between the access catheter **108** and the obstruction **2**. As discussed herein, in some variations, the access catheter **108** can be used to remove the obstruction **2** from the body once the obstruction is captured by a retrieval device. However, the greater the distance between the initial location of the obstruction **2** and the location of the access catheter **108**, the greater the risk that the obstruction **2** can break free from the retrieval device or become dislodged due to anatomic or environmental features, including but not limited to bifurcations, the wall of the lumen, the tortuousity of the anatomy, vessel wall plaque, etc.

FIG. **4**B illustrates an optional catheter **112** that advances from the access catheter **108** to the site of the obstruction **2**. Once at the site, the catheter **112** can deploy a retrieval device (not shown in FIG. **4**B) so that the retrieval device can engage the clot **2**. Alternatively, the catheter **112** can traverse the obstruction **2** as shown in FIG. **4**C and deploy a portion of the retrieval device **100** distally to the obstruction **2**. The physician then manipulates the retrieval device **100** to secure the obstruction **2**. For example, the physician can deploy the retrieval structure **200** distally to the obstruction **6** and withdraw the retrieval structure **200** proximally to secure the obstruction **2**. In another variation, the physician can position the retrieval structure **200** within the catheter **2** while the catheter **112** is through or adjacent to the obstruction **2**. Then, the physician can withdraw the catheter **112** to expose the retrieval structure **200** so that it secures to the obstruction **2** after expansion. In the illustrated example, the retrieval structure **200** comprises an elongated stent type structure that expands (or is expanded) to enmesh or secure to the obstruction. Although not illustrated, the system can include a distal capture filter or basket as described in any of the commonly assigned applications incorporated by reference herein.

Next, as shown in FIG. **4**E, the physician can further withdraw the catheter **112** to expose a cover **300** as described above. In many cases, the physician exposes the cove **300** once the retrieval structure **200** is engaged with the obstruction **2**. This sequential process allows for easier repositioning of the retrieval structure **200** if necessary. Alternatively, the cover **300** can be deployed prior to engaging the retrieval structure **200** with the obstruction **2**. If necessary, the physician can apply a proximal force on the delivery wire **202** while withdrawing the catheter **112** to prevent inadvertent movement of the obstruction **2** and retrieval device **200**.

FIG. **4**F illustrates the stage with a fully exposed the cover **300** and a catheter **112** moved closer towards the access sheath **108**. As shown, the free end **304** of the cover **300** is proximal to fixed end **302** of the cover **300**. As also noted above, the cover **300** can be a shape memory alloy that expands against the walls of the vessel **6** upon reaching body temperature. Alternatively, the cover **300** can be self expanding upon deployment into the vessel **6**. In some variations, the cover wall **306** comprises a porous material or construction that allows blood to continue to flow through the cover **300**.

In addition, some variations of the retrieval device **100** include a cover **300** that has at least a section that expands to a greater diameter or dimension than the retrieval structure **200**. This allows for expansion of the cover **300** against the wall of the vessel **6**. In most variation, expansion of the cover **300** provides sufficient friction against the walls of the vessel to overcome column strength of the cover walls **306** allowing for everting of the cover walls **306** over the retrieval structure **200** and obstruction **2** as discussed herein. As noted above, in alternate variations the cover **300** can expand a diameter or dimension that is equal to or less than the retrieval structure **200**.

FIG. **4**G illustrates proximal movement of the delivery wire **202**, which causes proximal translation of the obstruction **2** and retrieval structure **200**. Because the cover **300** is expanded against the walls of the vessel **6** the free end **304** of the cover **300** does not move or moves less than the fixed end **306** of the cover **300**. The fixed end **306** moves with the obstruction **2** and retrieval structure **200** in a proximal direction causing the cover walls **306** to evert over the obstruction **2** and retrieval structure **200**. Unlike a conventional funnel, the everting cover functions similar to a conveyor belt type movement as the obstruction and retrieval structure move together. This action allows for a passive type of protection since cover **300** does not need to be actuated over the obstruction **2** and retrieval structure **200** and can be performed in a quick manner by simply withdrawing the deployed retrieval device **100**.

FIG. **4**H illustrates a stage where the fixed end **306** of the cover **300** is now proximal to the free end **304**. As shown, the everted cover **300** forms a protective sheath or cover over the obstruction **2** and the retrieval structure **200**. FIG. **4**H also illustrates how the cover **300** protects the obstruction **2** and retrieval structure **200** as they are pulled along the vessel and navigate the tortuous anatomy, walls of the vessel, as well as bifurcations **8**. The cover **300** and cover wall **306** also protects the vasculature from the surface of the retrieval structure **200** and obstruction **2**.

FIG. **4**I shows the obstruction **2** and retrieval structure **200** protected by the cover **300** as the retrieval device **100** is positioned against or within the access catheter **108** in preparation for removal from the body. The retrieval device **100** can remain outside of the access catheter **108** as the physician removes both devices from the body. Alternatively, the cover **300** can assist in pulling the retrieval device **100** and obstruction **2** into the access catheter **108** by compressing the obstruction **2** as it is pulled into the access catheter **108**.

FIGS. **4**J and **4**K illustrate examples of an obstruction or other material **2** captured within a retrieval device **2** with a cover **300** further protecting the loaded retrieval device **200**.

FIGS. **5**A to **5**K show a variety of cover configurations. FIG. **5**A illustrates a retrieval device **100** having a retrieval structure **200** adjacent to a double layer cover **300** with an exterior wall **306** and an interior wall **308**.

FIG. **5**B shows a cover **300** with a free end **304** that tapers down about the delivery wire **202** where the cover **300** will eventually form a double wall configuration when the cover **300** everts over the retrieval structure **200**. The tapered free end **304** limits the cover **304** from moving once the retrieval structure **200** reaches the free end **304** thereby forming double wall protection over the retrieval structure **200**.

FIGS. **5**C and **5**D show how a fixed end **302** of a cover **300** can be pre-shaped to reduce the force required to evert the cover wall **306** or to lower the threshold to trigger passive covering of the retrieval structure by the cover. For example, as shown in FIG. **5**C, the fixed end **302** of the cover **300** can include a distal recess or opening **307**.

FIG. **5**E shows alternate variation of a passive cover **300** integrated into a retrieval device **100**. In this variation, the retrieval device **100** includes a control shaft or wire **202** to manipulate the working end of the retrieval device **100**. The cover **300** floats along the shaft **202** between two fixed anchors or nodes **220**, **222**. The cover **300** can float or slide between the fixed nodes **220**, **222**. The nodes **220**, **222** can comprise radiopaque marker bands, glue joints, or any other mechanical obstructions capable of stopping the translation of cover **300**. When the device **100** advances through a microcatheter, the rear or proximal node **220** limits rearward movement of the cover **300**. When positioned appropriately, the microcatheter can be withdrawn to expose the retrieval device **200** and cover **300** as described herein. When the retrieval structure **200** engages the obstruction (not shown) the retrieval device **100** can be withdrawn by pulling on the delivery shaft **202**. While this occurs, the cover **300**, being expanded against the vessel remains stationary (or moves at a slower rate than the obstruction and retrieval structure **200** due to the friction against the vessel wall). The retrieval structure **200** and clot enter the cover **300**, causing the distal node **222** to make contact with the near end **320** of the cover **300**. This contact causes the retrieval structure **200** and cover **300** to translate as an integrated unit. It should be appreciated that the cover could be a single layer or double layer cover, and could have any of the wire design variables and termination variables described herein.

FIG. **5**F illustrates a cover having a pre-set flattened cover wall **304** at a fixed end **302** that is spaced from a proximal end of the retrieval structure **200**. FIGS. **5**G to **5**I illustrate various layered covers **300**. The layered covers allow for shortening the axial length of the cover and therefore shortens the required translation length. Layering of the cover wall **306** allows for a shortened deployed length of the cover **300** when deployed in the vessel or body structure. As the cover **300** everts over the retrieval structure **200** the layered wall **306** extends. As a result, shortening the length reduces the length that the cover **300** extends into the proximal vessels and reduces the length of that the retrieval structure **200** must travel to become protected by the cover **300**. This also helps shorten the distance required to move the device **100** to complete eversion of the cover **300**.

FIG. **5**J shows a cover **300** that is constructed directly onto the retrieval structure **200** rather than the delivery shaft **202**. This construction also assists in reducing the distance necessary to complete passive protection of the retrieval structure by the cover.

FIG. **5**K show a variation of a cover **300** that is mounted in a distal direction over the retrieval device **200** and then everted in a proximal direction over the wires or shaft **202** as shown by arrows **230**. Once everted, as shown by FIG. **5**L, the device **100** is ready for deployment as discussed herein.

FIGS. **6**A to **6**B illustrate a variation of a cover **350** for use as describe herein. Additionally, the cover **350** can be used with any obstruction retrieval device not limited to the retrieval baskets and stents described herein. The covers **350** disclosed herein can be used where the physician desires to shield the obstruction being removed from the frictional effects of the arteries or from the local anatomy (e.g., branching vessels, tortuous anatomy, or other substances on the vessel walls). In use, the covers can be sized for use with guide catheters, micro-catheters, and/or distal access catheters. The covers can include any number of radiopaque marker bands to allow non-invasive imaging of the device (see marker **390** affixed between cover **350** and shaft **212** in FIG. **7**B as one example). In any case, once the retrieval device captures a clot or obstruction, as described above, the device and clot are protected by the cover so that the cover eliminates or reduces direct contact between the interior of the wall of the vessel and the clot.

FIGS. **6**A to **6**C show a variation in which a cover is created from one or more mesh tubes **372**. FIG. **6**B illustrates inversion of the tube **372** so that a first end **374** is drawn over the tube **372** towards a second end **376**. As shown in FIG. **6**C, this creates a double walled cover having an exterior wall **378** separated from an interior wall **380**. In one example, such a spacing or gap could range between 0.001 inches to 0.100 inches. However, any range is contemplated within alternative variations of the device. In some variations the inverted cover **350** is heat set to maintain a separation between layers or walls **378 380** of the cover **350**. Typically, if the cover **350** is not created from a radiopaque material, a marker band will be placed on the proximal end **376** and adjacent to a shaft or catheter to which the cover **350** is attached. In some variations the construction of the mesh material is compliant to allow for movement of a first part of the mesh relative to a second part of the mesh through compression and expansion of the mesh material. In such a case, the individual strands forming the mesh are moveable relative to one another to cause the mesh to be naturally compliant. Accordingly, this construction permits the inner wall **380** to move or deflect with the retrieval device and/or obstruction as the device is withdrawn into the cover **350**. In some variations, both ends of the mesh **374** and **376** are affixed to the catheter, shaft or wire.

In many variations, the cover mesh is selected to minimize friction when the interior layer **380** moves against the exterior layer **378**. For example, the braid pattern, wire, wire diameter, angle of the braid and or other features can be selected to reduce friction between the outer layer **378** and inner layer **380**. This permits the inner layer **380** to move proximally with a retrieval device while the outer layer remains stationary. Again, as discussed above, the construction of the mesh permits compression and expansion of the mesh layer to permit movement of the inner layer while the outer layer remains affixed when engaged against the vessel wall. In certain variations, the cover is heat set so that the inner layer has cushioning and the ability to deflect to assist in movement of the inner layer. FIG. **5**C also illustrates a cover **350** having a tapered design.

FIGS. **6**D to **6**L illustrate additional variations of cover construction to produce covers having more than two walls. For example, a mesh tube **372** is everted or drawn over a second end **376** in the direction **420**. As shown in FIG. **6**E this produces a dual layer cover having a open ends **422** and

424 and a folded end 426. The dual layer tube is then folded over again in the direction 420. This creates a cover construction with an exterior layer 378 and an interior layer 380 as well as a first intermediate layer 381 and a second intermediate layer 383. As shown in FIG. 6F, the cover can be set to assume the tapered shape having an opening at the first end 374 that is flared with the ends of the mesh at the second end 372, which are ultimately affixed to a shaft, wire or other catheter device as described herein.

FIG. 6G illustrates another example of a cover construction. As shown, a first mesh tube 372 is placed coaxially with a second tube 372. The concentric tubes are then everted in direction 420 to produce a four layer cover. As shown in FIG. 6H, the cover can comprise an interior mesh layer 380, and exterior mesh layer 378 as well as any number of intermediate layers 381, 383 depending on the number of tubes that are initially used. The second end 372 of the cover 350 includes four unconnected ends of the mesh tubes that can be affixed to a shaft or tube as discussed herein, while the first end 374 of the cover 350 can be shape set to taper from the opening.

FIGS. 6I to 6L illustrate another example of the construction of a multi-wall cover. As shown in FIG. 6I, a first end 374 of a mesh tube 372 is everted over and beyond a second end 376 in direction 420 to produce the configuration of FIG. 6J. Next, the first end 374 is everted or folded back in direction 420 to produce the configuration of FIG. 6K. Finally, the first end 374 is folded again in direction 420 so that the ends 374 and 376 are even to produce the cover configuration shown in FIG. 6K. Again, one end of the cover 350 can be set to form the tapered shape while the other respective end can be affixed to a catheter or shaft.

Although the covers of the present disclosure are presented without additional structures, it should be noted that these covers are coupled with a shaft or other member so that the cover can be advanced within the target anatomy to assist in removal of a device, structure, or debris from the site.

FIGS. 7A to 7C show addition variations of covers 350. FIG. 7A illustrates a cover in which the cover wall as defined by the inner layer 380 and outer layer 378 is set in a shape that varies along a length of the cover. For example, the end adjacent to the cover opening 382 can be set to a bulbous shape. Such a configuration assists in maintaining separation of layers 378 and 380, which aids in re-entry of the retrieval device. Additional configurations of cover walls that vary in thickness are within the scope of this disclosure.

One of the benefits of using a cover 350 as described herein is that the cover reduces flow through the vessel when deployed so that the retrieval device can remove the obstruction without the full force of the flow of blood opposing the obstruction. Typically, conventional devices relied upon the use of an inflated balloon to obstruct flow. However, use of a cover eliminates the need for total occlusion of blood flow. FIG. 7B illustrates a further improvement on a cover 350 that aids in flow reduction. As shown, the cover 350 includes a dense region 386 and a relatively less dense region 384. This configuration permits greater blood flow through the region 385 while region 386 reduces or prevents blood flow. Furthermore, the distal section of the cover is more flexible and conformable. Additional mesh layers can be added to any of the cover designs to alter flow characteristics or even provide reinforcement to the cover. Alternatively, or in combination, the braid density can be altered to adjust the porosity of the braid at different sections. Furthermore, additional braid layers can also be used to affect porosity of portions of the cover or even the entire cover. Deployment of a cover can reduce blood flow by 30% to 40%. Adding additional layers or coatings can additionally reduce flow.

FIG. 7C shows another variation of a cover 350 in which the mesh partially or totally is obscured using a polymeric coating 388 that reduces the permeability of the mesh design. Furthermore, drugs or other substances can be placed within the cover wall of any of the covers or can be deposited on the cover using the polymeric coatings. In some examples, the covers described herein can range from a length of 10 mm up to 50 mm. The OD at the opening of the cover can range from 7 mm and could range between 4 mm to 10 mm. Again, any range of dimensions is contemplated within the disclosure.

The covers described herein can further be stacked on a device. For example, two or more covers can be placed on a device to provide added protection.

The cover/rentry devices described herein can be constructed of any material currently used in vascular applications, including those discussed above. Furthermore, fabrication of the cover from a DFT material can provide additional benefits as the entire cover remains radiopaque and can be imaged non-invasively. Furthermore, the covers can be provided with any type of medicament or bioactive substance either in a polymer that coats the mesh or in a delivery agent within the mesh or between layers. Such substances include tpa, urokinase, IIb/IIIa inhibitors, and other clot disruptors or inhibitors.

FIG. 8 illustrates another variation of a retrieval device 400 including a distal capture portion 426 coupled to one or more leading wires in the form of a main bundle 402. The main bundle extends through a sheath 106 that includes a proximal capture portion 460. The configuration of the retrieval device 400 can incorporate the proximal and distal capture portions discussed herein as well as various other configurations discussed in the commonly assigned patent applications noted above.

An end 464 of the proximal capture portion 460 is affixed to a distal end of the sheath 106. However, as noted above, other variations are within the scope of the disclosure. The main bundle 402 can optionally terminate at a handle 442. As noted above, in certain variations, the main bundle is joined to a stiffer wire or stiffer bundle of wires. This allows the device 400 to have a very flexible distal section with a relatively stiffer proximal section. The device 400 can have a proximal bundle 403 that comprises either the exposed wires or a covering/tube over the wires. In certain variations, the bundle or wire 402, 403 can be encapsulated with a coating. The device also includes a cover 300 adjacent to the retrieval device.

The proximal end of the sheath 106 includes a sheath handle 444. As discussed herein, axial movement of the bundle 402 or proximal bundle 403 (typically at the handle 442) results in movement 126, or translation of the bundle within the sheath 106. This action moves the distal capture portion 426 (as shown by arrows 126). In certain variations, the device 400 is loaded into a microcatheter (not shown but discussed above) that is delivered to the site of the obstruction and crosses the obstruction.

In some variations, the sheath hub 444 includes one or more locking hubs 446. Where actuation (either axial or rotational) of the locking hub 446 locks the main bundle 402 relative to the sheath handle 444 and sheath 106. It follows that such locking action also locks the distal capture portion 426 relative to the proximal capture portion 460. A variety of methods can be employed to increase a frictional interference between the locking hub 446 and the proximal bundle 403. As a result, when a physician determines a length of an obstruction, the physician can set a spacing between the capturing portions **426 460** by locking the proximal bundle **403** relative to the sheath hub **444**. Accordingly, the proximal bundle **403** can include any type of incremental markings to allow the physician to readily determine a spacing of the capturing portions. As illustrated, the sheath hub **444** can include additional injection ports to deliver fluid or other substances through the sheath **106**.

As noted above, the device **400** can be used with a micro-catheter. In those variations it is important that the device **400** is loaded without damaging the distal bundle **402**, capture portions **426 460**, and/or sheath **106**. As a result, the device **400** can include an optional cover **486** that reduces the proximal capture portion **460** (and/or the distal capture portion **426**) for loading within the microcatheter and/or sheath **106**.

Another variation of the device **400** includes an insertion tool **480** slidably affixed to the sheath **480**. Because variations of the device **400** can be extremely flexible, the insertion tool **480** can be used to provide column strength to the sheath **106**, bundle **402** or other components as the device **400** is pushed into the microcatheter. The insertion tool comprises a rigid section **482** and a frictional coupler **484**. The rigid section **282** has a column strength that supports the device **400** to prevent buckling. The frictional coupler **484** can be a flexible material that allows an operator to squeeze or grip the coupler **484** to create a temporary frictional interface between the loading tool **480** and the device **400** (typically the sheath **106**). Such an action allows axial advancement of the device **400** as the loading tool **480** is advanced into the microcatheter. Once the rigid section **482** is fully inserted into the microcatheter, the operator releases the frictional coupler **484** and can withdraw the loading tool **480** from the catheter without withdrawing the device **400**. The insertion tool **480** can also include an optional loading tube **486** slidably coupled to the rigid section **482**. When used, the cover **486** can withdraw the proximal and distal capturing portion **226 260** within the loading tube **486**. The loading tube **486** then couples to a microcatheter allowing the capturing portions to advance therein as the rigid section **482** and frictional coupler **484** advance the device **400** relative to the loading tube **486**.

FIGS. **9**A to **9**C show cross sectional views taken along the line **9**A-**9**A in FIG. **2**A. As shown, the wire form construction described herein allows for a number of configurations depending on the particular application. For example, the individual wires **254** (as discussed herein) may themselves comprise a bundle of smaller wires or filaments. In addition, the wires can be selected from materials such as stainless steel, titanium, platinum, gold, iridium, tantalum, Nitinol, alloys, and/or polymeric strands. In addition, the wires used in a device may comprise a heterogeneous structure by using combinations of wires of different materials to produce a device having the particular desired properties. For example, one or more wires in the device may comprise a shape memory or superelastic alloy to impart predetermined shapes or resiliency to the device. In some variations, the mechanical properties of select wires can be altered. In such a case, the select wires can be treated to alter properties including: brittleness, ductility, elasticity, hardness, malleability, plasticity, strength, and toughness.

The device may include a number of radiopaque wires, such as gold and platinum for improved visibility under fluoroscopic imaging. In other words, any combination of materials may be incorporated into the device. In addition to the materials, the size of the wires may vary as needed. For example, the diameters of the wires may be the same or may vary as needed.

In addition, the individual wires may have cross-sectional shapes ranging from circular, oval, d-shaped, rectangular shape, etc. FIG. **9**A illustrates one possible variation in which a number of circular wires **254** are included around another larger wire **256**. Moreover, the device is not limited to having wires having the same cross-sectional shape or size. Instead, the device can have wires having different cross-sectional shapes. For example, as shown in FIG. **9**B, one or more wires **256** can have a different cross-sectional shape or size than a reminder of the wires **254**. Clearly, any number of variations is within the scope of this disclosure. This construction can apply to the retrieval portion, capturing portion and/or the covering portion of the device.

To illustrate one such example, a device can have 8-12 wires made of 0.003" round superelastic material (e.g., Nitinol). The device may additionally have 2-4 wires made from 0.002" platinum for fluoroscopy. Of the 8-12 Nitinol wires, 1-4 of these wires can be made of a larger diameter or different cross-section to increase the overall strength of the device. Finally, a couple of polymer fibers can be added where the fibers have a desired surface property for clot adherence, etc. Such a combination of wires provides a composite device with properties not conventionally possible in view of other formation means (such as laser cutting or etching the shape from a tube or joining materials with welds, etc.). Clearly, any number of permutations is possible given the principles of the invention.

In another example, the device may be fabricated from wires formed from a polymeric material or composite blend of polymeric materials. The polymeric composite can be selected such that it is very floppy until it is exposed to either the body fluids and or some other delivered activator that causes the polymer to further polymerize or stiffen for strength. Various coatings could protect the polymer from further polymerizing before the device is properly placed. The coatings could provide a specific duration for placement (e.g., 5 minutes) after which the covering degrades or is activated with an agent (that doesn't affect the surrounding tissues) allowing the device to increase in stiffness so that it doesn't stretch as the thrombus is pulled out. For example, shape memory polymers would allow the device to increase in stiffness.

In another variation, one or more of the wires used in the device may comprise a Drawn Filled Tube (DFT) such as those provided by Fort Wayne Metals, Fort Wayne, Ind. As shown in FIG. **9**C, such a DFT wire **252** comprises a first material or shell **258** over a second material **260** having properties different from the outer shell. While a variety of materials can be used, one variation under the present devices includes a DFT wire having a superelastic (e.g., Nitinol) outer tube with a radiopaque material within the super-elastic outer shell. For example, the radiopaque material can include any commercially used radiopaque material, including but not limited to platinum, iridium, gold, tantalum, or similar alloy. One benefit of making a capturing portion from the DFT wire noted above, is that rather than having one or more markers over the capturing portion, the entire capturing portion can be fabricated from a super-elastic material while, at the same time, the super-elastic capturing portion is made radiopaque given the core of radiopaque material within the super-elastic shell. Clearly, any composite DFT wire **252** can be incorporated into the system and capturing portions described herein.

Another aspect applicable to all variations of the devices is to configure the devices or portions thereof that engage the obstruction to improve adherence to the obstruction. One such mode includes the use of coatings that bond to certain clots (or other materials causing the obstruction.) For example, the wires may be coated with a hydrogel or adhesive that bonds to a thrombus. Accordingly, as the device secures about a clot, the combination of the additive and the mechanical structure of the device may improve the effectiveness of the device in removing the obstruction. Coatings may also be combined with the capturing portions or catheter to improve the ability of the device to encapsulate and remove the obstruction (e.g., a hydrophilic coating).

Such improvements may also be mechanical or structural. Any portion of the capturing portion can have hooks, fibers, or barbs that grip into the obstruction as the device surrounds the obstruction. The hooks, fibers, or barbs **370** can be incorporated into any portion of the device. However, it will be important that such features do not hinder the ability of the practitioner to remove the device from the body.

In addition to additives, the device can be coupled to an RF or other power source (such as **14** or **16** in FIG. **1**), to allow current, ultrasound or RF energy to transmit through the device and induce clotting or cause additional coagulation of a clot or other the obstruction.

FIGS. **10**A to **10**E illustrate additional variations of covers **300** for use as described above. For example, as show in FIG. **10**A, a cover **300** can comprise a single wire, coil, or laser cut tube **351**. Alternatively, as shown in FIG. **10**B, the cover **300** can comprises two or more **351**, **352** wires or coils. FIG. **10**C shows a cover **300** comprising a coil **351** inside a mesh structure **354**. A variation of the device shown in FIG. **10**C can include a compliant atraumatic mesh **354** that is radially supported by the coil (whether interior or exterior to the mesh). The coil **351** provides the outward force against the vessel. FIG. **10**D illustrates a polymeric film or membrane **356** coupled to a coil **351**. The polymeric film **356** can be permeable to fluid flow or impermeable. FIG. **10**E illustrates a dual layer braid construction having an inner braid **358** and an outer braid **360**. The braids can be constructed to have unique properties. For example, the inner braid **358** can be composed of fewer wires or larger diameter wires, such that it provides an expansion force against the vessel wall. The outer braid **360** can comprise a softer construction and increased compliance. Accordingly, it can be comprised of a number of smaller diameter wires having a denser pattern to provide increased surface area to protect the obstruction as it is removed from the body. Alternatively, these two constructional elements (e.g., braids of varying diameters) can be combined into a single layer or even multiple layers for the cover.

FIG. **11**A illustrates yet another variation of a device **100** having a retrieval structure **200** and cover **300** where the cover is simply fabricated from the same material as the retrieval structure so long as it functions as described herein. The variation can optionally include one or more barbs **370** to increase resistance against a vessel wall.

FIGS. **11**B and **11**C illustrate a variation where the cover **300** comprises a balloon material. FIG. **11**B illustrates the balloon cover **370** prior to deployment. FIG. **11**C illustrates the balloon cover **370** once deployed.

The retrieval devices described herein can optionally comprise elongated stents **400** as shown in FIGS. **12**A to **12**E. These stents **400** can include any number of features to better assist the stent **400** in becoming enmeshed into the obstruction. For example, FIG. **12**A illustrates a variation of a stent **400** affixed to a shaft **412**. As noted herein, the shaft **412** can include a lumen extending therethrough. Alternatively, the shaft **412** can include a solid member with the stent **400** affixed to a distal end thereof. The variation shown in FIG. **12**A includes a stent where a distal end **414** that is "closed off" by intersecting elements or wires **402 403**. Accordingly, any of the variations of the stents disclosed herein can include an open lumen type stent or a closed lumen type stent as shown in FIG. **12**A. As noted herein, the wires forming the stent **400** can comprise a single wire that is wound from a first direction (e.g., from proximal to distal) and then wound back in a second direction (e.g., from distal to proximal).

FIG. **12**A also illustrates a stent **400** comprised of twisted wires **402** or elements. For example, FIG. **12**B shows a magnified view of the section **12**B in FIG. **12**A. As illustrated, the elements **402 403** are twisted to increase the surface area at the exterior perimeter of the stent **400**. The twisting or spiraling of the elements **402 403** creates additional surface area to increase the ability of the stent **400** to capture debris, thrombus, foreign body, etc. as the stent is expanded against the debris. The twisting elements **402 403** can twist along the entire length of the stent **400** or along one or more portions of the stent. In certain variations, the twisting of the elements **402 403** is sufficiently loose such that as the stent expands into a clot or obstruction, the twisted pairs slightly separate to allow material to become trapped between the elements making up the pairs. The construction shown in FIGS. **12**A and **12**B also provide an additional benefit to a retrieval stent. In the illustrated variation, the twisted or spiraling elements interlock with crossing elements to form intersections **405** that provided added radial expansive force. As shown, a first twisted element **407** passes in between elements **402 403** of an intersecting element **409**. When in an expanded state, the element on the interior of the intersection **405** (in this case element **403**) provides an added outward radial force against the intersection **405**. However, since the elements are not affixed but instead are slidable at the intersection **405**, the force required to linearlize and compress the stent **400** is reduced due to the fact that the intersections are not affixed but slidable over the adjacent elements. This reduced linearization force allows the stent to be compressed to a small diameter for positioning within a microcatheter but allow for a significant radial expansive force once removed from the microcatheter. This design allows for a reduction in radial force of the stent against the vessel wall when the stent is pulled and removed from the vessel. However, this design also provides a high degree of radial force due to the interweaving of elements when the stent is deployed in the vessel prior to withdrawal of the stent.

FIGS. **12**C to **12**F illustrate another variation of types of stents **400** that have an irregular surface at an exterior of the stent **400** that is formed by an intersection of elements **402 403**. The intersection or crossing of the elements forms a type of barb or knuckle **416** that creates an irregular surface on the exterior of the stent **400**. FIG. **12**C illustrates a variation of a stent **400** having a plurality of knuckles **52** that are radially spaced about an axis **390** of the stent **400**. FIG. **12**E shows another variation of a stent **400** with knuckles **416** aligned with an axis **390** of the stent **400** as shown in FIG. **12**D. Although the figures show the axial and radial aligned knuckles **416** on separate devices, both types of knuckles **416** can be incorporated into a single stent structure. Varying the alignment of knuckles can permit increased radial force as the stent expands into the obstruction or increased flexibility as the stent navigates through tortuous anatomy.

FIG. **12**G illustrates a proximal end of the stent structure **400** as shown, a plurality of elements **402** and **403** extend along the shaft **412** and diverge to form the fluid permeable closed proximal end of the stent structure **400**. The elements **402** and **403** that extend along the shaft **412** can be covered by a sheath, tube, spiral cut tube, or any structure **418** that prevents separation of the elements **402 403**. A variation of the stent structure **402** includes a construction where the elements **402 403** are not glued, welded, or have any similar type of joint in the distal portion **420** of the shaft **412**. Instead, the joint **411** is located proximal to the distal section of the shaft **412** in an intermediate section **422**. Because joints or other similar features reduce flexibility of the joined structure, positioning the joints **411** in a proximal area allows the distal portion **420** of the shaft to remain flexible.

The methods described herein may also include treating the obstruction prior to attempting to remove the obstruction. Such a treatment can include applying a chemical or pharmaceutical agent with the goal of making the occlusion shrink or to make it more rigid for easier removal. Such agents include, but are not limited to chemotherapy drugs, or solutions, a mild formalin, or aldehyde solution.

As for other details of the present invention, materials and manufacturing techniques may be employed as within the level of those with skill in the relevant art. The same may hold true with respect to method-based aspects of the invention in terms of additional acts that are commonly or logically employed. In addition, though the invention has been described in reference to several examples, optionally incorporating various features, the invention is not to be limited to that which is described or indicated as contemplated with respect to each variation of the invention.

Various changes may be made to the invention described and equivalents (whether recited herein or not included for the sake of some brevity) may be substituted without departing from the true spirit and scope of the invention. Also, any optional feature of the inventive variations may be set forth and claimed independently, or in combination with any one or more of the features described herein. Accordingly, the invention contemplates combinations of various aspects of the embodiments or combinations of the embodiments themselves, where possible. Reference to a singular item, includes the possibility that there are plural of the same items present. More specifically, as used herein and in the appended claims, the singular forms "a," "and," "said," and "the" include plural references unless the context clearly dictates otherwise.

It is important to note that where possible, aspects of the various described embodiments, or the embodiments themselves can be combined. Where such combinations are intended to be within the scope of this disclosure.

We claim:

1. A device for deployment within a blood vessel using a catheter, the device comprising:

a shaft having a distal end and a proximal end;

a braided cover having a free and open distal section and a proximal end coupled to the shaft at a location proximal to the distal end of the shaft, wherein the proximal end comprises at least one peak pointing in a proximal direction and at least one valley pointing in a distal direction, wherein the braided cover is configured to self-expand against the vessel wall when deployed from the catheter; and

an expandable helical coils within the braided cover and configured to conform to a profile of the braided cover.

2. The device of claim **1**, wherein the braided cover comprises a nitinol braid.

3. The device of claim **1**, wherein the braided cover is formed from a mesh tube.

4. The device of claim **1**, wherein the braided cover permits continued blood flow through the blood vessel when self-expanded against the vessel wall.

5. The device of claim **1**, further comprising a node positioned on the shaft between the distal end and the proximal end.

6. The device of claim **5**, wherein the node is configured to obstruct movement of the braided cover relative to the shaft.

7. The device of claim **5**, wherein the node comprises a radiopaque marker band.

8. The device of claim **5**, wherein the node is a first node, and the device further comprises a second node positioned on the shaft between the distal end and the proximal end, the second node being spaced apart from the first node.

9. The device of claim **1**, further comprising a radiopaque marker at the proximal end of the braided cover.

10. The device of claim **1**, wherein the helical coil is radiopaque.

11. The device of claim **1**, wherein the helical coil is a single coil configured to provide radial support to the braided cover.

12. A device for deployment within a blood vessel using a catheter, the device comprising:

a shaft having a distal end and a proximal end; and

a cover configured to self-expand into contact with the vessel wall when deployed from the catheter, the cover including:

a mesh structure having a free and open distal section and a proximal end coupled to the shaft at a location proximal to the distal end of the shaft, wherein the proximal end comprises at least one peak pointing in a proximal direction and at least one valley pointing in a distal direction, and

an expandable helical coils inside the mesh structure and configured to conform to a profile of the mesh structure.

13. The device of claim **12**, wherein the mesh structure comprises a braid.

14. The device of claim **12**, wherein the mesh structure comprises a tube.

15. The device of claim **12**, wherein the cover permits continued blood flow through the blood vessel when self-expanded against the vessel wall.

16. The device of claim **12**, further comprising at least one anchor coupled to the shaft between the distal end portion and the proximal end portion.

17. The device of claim **16**, wherein the at least one anchor is configured to constrain translation of the cover along the shaft.

18. The device of claim **16**, wherein the at least one anchor comprises a radiopaque marker band.

19. The device of claim **16**, wherein the at least one anchor comprises a distal anchor and a proximal anchor.

20. The device of claim **12**, further comprising a radiopaque marker at the proximal end of the cover.

21. The device of claim **12**, wherein the helical coil is radiopaque.

22. The device of claim **12**, wherein the helical coil is a single coil configured to provide radial support to the mesh structure.

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